

Exercise-3 : Matching Type Problems

1.

	Column-I		Column-II
(A)	If ${}^{n-1}C_r = (k^2 - 3){}^nC_{r+1}$ and $k \in \mathbb{R}^+$, then least value of $5[k]$ is (where $[\cdot]$ represents greatest integer function)	(P)	10
(B)	$\sum_{i=0}^m {}^{20}C_i {}^{40}C_{m-i}$, where ${}^nC_r = 0$ if $r > n$, is maximum when $\frac{m}{5}$ is	(Q)	5
(C)	Number of non-negative integral solutions of inequation $x + y + z \leq 4$ is	(R)	35
(D)	Let $A = \{1, 2, 3, 4, 5\}$, $f : A \rightarrow A$, The number of onto functions such that $f(x) = x$ for atleast 3 distinct $x \in A$, is not a multiple of	(S)	6
		(T)	12

2.

	Column-I		Column-II
(A)	Number of real solution of $(x^2 + 6x + 7)^2 + 6(x^2 + 6x + 7) + 7 = x$ is/are	(P)	15
(B)	If $P = \sum_{r=0}^n {}^nC_r$; $q = \sum_{r=0}^m {}^mC_r$ $(15)^r$ ($m, n \in \mathbb{N}$) and if $P = q$ and m, n are least then $m + n =$	(Q)	5
(C)	Remainder when $1! + 3! + 5! + \dots + 2011!$ is divided by 56 is	(R)	3
(D)	Inequality $\left 1 - \frac{ x }{1+ x } \right \geq \frac{1}{2}$ holds for x , then number of integral values of 'x' is/are	(S)	0

3. Match the following

	Column-I		Column-II
(A)	If the sum of first 84 terms of the series $\frac{4+\sqrt{3}}{1+\sqrt{3}} + \frac{8+\sqrt{15}}{\sqrt{3}+\sqrt{5}} + \frac{12+\sqrt{35}}{\sqrt{5}+\sqrt{7}} + \dots$ is $549k$, then k is equal to	(P)	3

(B)	If $x, y \in \mathbb{R}$, $x^2 + y^2 - 6x + 8y + 24 = 0$, the greatest value of $\frac{16}{5} \cos^2(\sqrt{x^2 + y^2}) - \frac{24}{5} \sin(\sqrt{x^2 + y^2})$ is	(Q)	2
(C)	If $(\sqrt{3} + 1)^6 + (\sqrt{3} - 1)^6 = 416$, if $xyz = [(\sqrt{3} + 1)^6]$, $x, y, z \in \mathbb{N}$, (where $[\cdot]$ denotes the greatest integer function), then the number of ordered triplets (x, y, z) is	(R)	5
(D)	If $(1+x)(1+x^2)(1+x^4) \dots (1+x^{128}) = \sum_{r=0}^n x^r$, then $\frac{n}{85}$ is equal to	(S)	9

Answers

- $A \rightarrow Q; B \rightarrow S; C \rightarrow R; D \rightarrow P, Q, R, S, T$
- $A \rightarrow S; B \rightarrow Q; C \rightarrow P; D \rightarrow R$
- $A \rightarrow Q; B \rightarrow R; C \rightarrow S; D \rightarrow P$

Exercise-4 : Subjective Type Problems

1. The sum of the series $3 \cdot {}^{2007}C_0 - 8 \cdot {}^{2007}C_1 + 13 \cdot {}^{2007}C_2 - 18 \cdot {}^{2007}C_3 + \dots$ upto 2008 terms is K , then K is :
2. In the polynomial function $f(x) = (x-1)(x^2-2)(x^3-3)\dots(x^{11}-11)$ the coefficient of x^{60} is :
3. If $\sum_{r=0}^{3n} a_r (x-4)^r = \sum_{r=0}^{3n} A_r (x-5)^r$ and $a_k = 1 \forall k \geq 2n$ and $\sum_{r=0}^{3n} d_r (x-8)^r = \sum_{r=0}^{3n} B_r (x-9)^r$ and $\sum_{r=0}^{3n} d_r (x-12)^r = \sum_{r=0}^{3n} D_r (x-13)^r$ and $d_k = 1 \forall k \geq 2n$. The find the value of $\frac{A_{2n} + D_{2n}}{B_{2n}}$.
4. If $3^{101} - 2^{100}$ is divided by 11, the remainder is
5. Find the hundred's digit in the co-efficient of x^{17} in the expansion of $(1 + x^5 + x^7)^{20}$.
6. Let $x = (3\sqrt{6} + 7)^{89}$. If $\{x\}$ denotes the fractional part of ' x ' then find the remainder when $x\{x\} + (x\{x\})^2 + (x\{x\})^3$ is divided by 31.
7. Let $n \in N$; $S_n = \sum_{r=0}^{3n} ({}^{3n}C_r)$ and $T_n = \sum_{r=0}^n ({}^{3n}C_{3r})$. Find $|S_n - 3T_n|$.
8. Find the sum of possible real values of x for which the sixth term of $\left(3^{\log_3 \sqrt{9|x-2|}} + 7^{\frac{1}{5} \log_7 (3^{x-2}-9)} \right)^7$ equal 567 :
9. Let q be a positive integer with $q \leq 50$.
If the sum ${}^{98}C_{30} + 2 \cdot {}^{97}C_{30} + 3 \cdot {}^{96}C_{30} + \dots + 68 \cdot {}^{31}C_{30} + 69 \cdot {}^{30}C_{30} = {}^{100}C_q$
Find the sum of the digits of q .
10. The remainder when $\left(\sum_{k=1}^5 {}^{20}C_{2k-1} \right)^6$ is divided by 11, is :
11. Let $a = 3^{\frac{1}{223}} + 1$ and for all $n \geq 3$, let
 $f(n) = {}^nC_0 \cdot a^{n-1} - {}^nC_1 \cdot a^{n-2} + {}^nC_2 \cdot a^{n-3} - \dots + (-1)^{n-1} {}^nC_{n-1} \cdot a^0$.
If the value of $f(2007) + f(2008) = 3^7 k$ where $k \in N$ then find k
12. In the polynomial $(x-1)(x^2-2)(x^3-3)\dots(x^{11}-11)$, the coefficient of x^{60} is :
13. Let the sum of all divisors of the form $2^p \cdot 3^q$ (with p, q positive integers) of the number $19^{88} - 1$ be λ . Find the unit digit of λ .

14. Find the sum of possible real values of x for which the sixth term of $\left(3^{\log_3 \sqrt{9^{x-2}}} + 7^{\left(\frac{1}{5}\right) \log_7 (3^{x-2}-9)}\right)^7$ equals 567.
15. Let $1 + \sum_{r=1}^{10} (3^r \cdot {}^{10}C_r + r \cdot {}^{10}C_r) = 2^{10} (\alpha \cdot 4^5 + \beta)$ where $\alpha, \beta \in N$ and $f(x) = x^2 - 2x - k^2 + 1$. If α, β lies between the roots of $f(x) = 0$. Then find the smallest positive integral value of k .
16. Let $S_n = {}^nC_0 {}^nC_1 + {}^nC_1 {}^nC_2 + \dots + {}^nC_{n-1} {}^nC_n$ if $\frac{S_{n+1}}{S_n} = \frac{15}{4}$; find the sum of all possible values of n ($n \in N$)

Answers

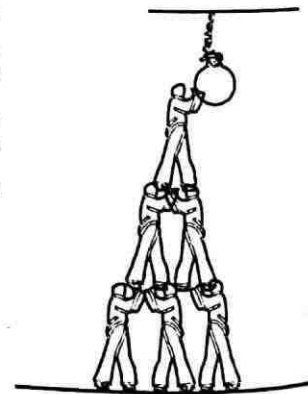
1.	0	2.	1	3.	2	4.	2	5.	4	6.	0	7.	2
8.	4	9.	5	10.	3	11.	9	12.	(1)	13.	(4)	14.	(4)
15.	5	16.	6										



Exercise-1 : Single Choice Problems

- The boy comes from a family of two children; What is the probability that the other child is his sister ? :
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{4}$
- If A be any event in sample space then the maximum value of $3\sqrt{P(A)} + 4\sqrt{P(\bar{A})}$ is :
 (a) 4 (b) 2
 (c) 5 (d) Can not be determined
- Let A and B be two events, such that $P(\bar{A} \cup B) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{1}{4}$, where $\bar{A}$ stands for complement of event A . Then events A and B are :
 (a) equally likely and mutually exclusive (b) equally likely but not independent
 (c) independent but not equally likely (d) mutually exclusive and independent
- Let n ordinary fair dice are rolled once. The probability that at least one of the dice shows an odd number is $\left(\frac{31}{32}\right)$ than ' n ' is equal to :
 (a) 3 (b) 4 (c) 5 (d) 6
- Three a 's, three b 's and three c 's are placed randomly in a 3×3 matrix. The probability that no row or column contain two identical letters can be expressed as $\frac{p}{q}$, where p and q are coprime then $(p + q)$ equals to :
 (a) 151 (b) 161 (c) 141 (d) 131
- A set contains $3n$ members. Let P_n be the probability that S is partitioned into 3 disjoint subsets with n members in each subset such that the three largest members of S are in different subsets. Then $\lim_{n \rightarrow \infty} P_n =$
 (a) $\frac{2}{7}$ (b) $\frac{1}{7}$ (c) $\frac{1}{9}$ (d) $\frac{2}{9}$

7. Three different numbers are selected at random from the set $A = \{1, 2, 3, \dots, 10\}$. Then the probability that the product of two numbers equal to the third number is $\frac{p}{q}$, where p and q are relatively prime positive integers then the value of $(p + q)$ is :
 (a) 39 (b) 40 (c) 41 (d) 42
8. Mr. A's T.V. has only 4 channels ; all of them quite boring so he naturally desires to switch (change) channel after every one minute. The probability that he is back to his original channel for the first time after 4 minutes can be expressed as $\frac{m}{n}$; where m and n are relatively prime numbers. Then $(m + n)$ equals :
 (a) 27 (b) 31 (c) 23 (d) 33
9. Letters of the word TITANIC are arranged to form all the possible words. What is the probability that a word formed starts either with a T or a vowel ?
 (a) $\frac{2}{7}$ (b) $\frac{4}{7}$ (c) $\frac{3}{7}$ (d) $\frac{5}{7}$
10. A mapping is selected at random from all mappings $f : A \rightarrow A$ where set $A = \{1, 2, 3, \dots, n\}$. If the probability that mapping is injective is $\frac{3}{32}$, then the value of n is :
 (a) 3 (b) 4 (c) 8 (d) 16
11. A 4 digit number is randomly picked from all the 4 digit numbers, then the probability that the product of its digit is divisible by 3 is :
 (a) $\frac{107}{125}$ (b) $\frac{109}{125}$
 (c) $\frac{111}{125}$ (d) None of these
12. To obtain a gold coin; 6 men, all of different weight, are trying to build a human pyramid as shown in the figure. Human pyramid is called "stable" if some one not in the bottom row is "supported by" each of the two closest people beneath him and no body can be supported by anybody of lower weight. Formation of 'stable' pyramid is the only condition to get a gold coin. What is the probability that they will get gold coin ?
 (a) $\frac{1}{45}$ (b) $\frac{2}{45}$
 (c) $\frac{4}{45}$ (d) $\frac{1}{5}$
13. From a pack of 52 playing cards; half of the cards are randomly removed without looking at them. From the remaining cards, 3 cards are drawn randomly. The probability that all are king.



(a) $\frac{1}{(25)(17)(13)}$

(b) $\frac{1}{(25)(15)(13)}$

(c) $\frac{1}{(52)(17)(13)}$

(d) $\frac{1}{(13)(51)(17)}$

14. A bag contains 10 white and 3 black balls. Balls are drawn one by one without replacement till all the black balls are drawn. The probability that the procedure of drawing balls will come to an end at the seventh draw is :

(a) $\frac{15}{286}$

(b) $\frac{105}{286}$

(c) $\frac{35}{286}$

(d) $\frac{7}{286}$

15. Let S be the set of all function from the set $\{1, 2, \dots, 10\}$ to itself. One function is selected from S , the probability that the selected function is one-one onto is :

(a) $\frac{9!}{10^9}$

(b) $\frac{1}{10}$

(c) $\frac{100}{10!}$

(d) $\frac{9!}{10^{10}}$

16. Two friends visit a restaurant randomly during 5 pm to 6 pm. Among the two, whoever comes first waits for 15 min and then leaves. The probability that they meet is :

(a) $\frac{1}{4}$

(b) $\frac{1}{16}$

(c) $\frac{7}{16}$

(d) $\frac{9}{16}$

17. Three numbers are randomly selected from the set $\{10, 11, 12, \dots, 100\}$. Probability that they form a Geometric progression with integral common ratio greater than 1 is :

(a) $\frac{15}{{}^{91}C_3}$

(b) $\frac{16}{{}^{91}C_3}$

(c) $\frac{17}{{}^{91}C_3}$

(d) $\frac{18}{{}^{91}C_3}$

Answers

1.	(a)	2.	(c)	3.	(c)	4.	(c)	5.	(c)	6.	(d)	7.	(c)	8.	(b)	9.	(d)	10.	(b)
11.	(a)	12.	(a)	13.	(a)	14.	(a)	15.	(a)	16.	(c)	17.	(d)						

Exercise-2 : One or More than One Answer is/are Correct

- A consignment of 15 record players contain 4 defectives. The record players are selected at random, one by one and examined. The one examined is not put back. Then :

 - Probability of getting exactly 3 defectives in the examination of 8 record players is $\frac{{}^4C_3 {}^{11}C_5}{{}^{15}C_8}$.
 - Probability that 9th one examined is the last defective is $\frac{8}{197}$.
 - Probability that 9th examined record player is defective, given that there are 3 defectives in first 8 players examined is $\frac{1}{7}$.
 - Probability that 9th one examined is the last defective is $\frac{8}{195}$.
- If $A_1, A_2, A_3, \dots, A_{1006}$ be independent events such that $P(A_i) = \frac{1}{2^i}$ ($i = 1, 2, 3, \dots, 1006$) and probability that none of the events occurs be $\frac{\alpha!}{2^\alpha (\beta!)^2}$, then :

 - β is of form $4k + 2, k \in I$
 - $\alpha = 2\beta$
 - β is a composite number
 - α is of form $4k, k \in I$
- A bag contains four tickets marked with 112, 121, 211, 222 one ticket is drawn at random from the bag. let E_i ($i = 1, 2, 3$) denote the event that i^{th} digit on the ticket is 2. Then :

 - E_1 and E_2 are independent
 - E_2 and E_3 are independent
 - E_3 and E_1 are independent
 - E_1, E_2, E_3 are independent
- For two events A and B let, $P(A) = \frac{3}{5}, P(B) = \frac{2}{3}$, then which of the following is/are correct ?

 - $P(A \cap \bar{B}) \leq \frac{1}{3}$
 - $P(A \cup B) \geq \frac{2}{3}$
 - $\frac{4}{15} \leq P(A \cap B) \leq \frac{3}{5}$
 - $\frac{1}{10} \leq P(\bar{A}/B) \leq \frac{3}{5}$

Answers

1.	(a, c, d)	2.	(a, b, c, d)	3.	(a, b, c)	4.	(a, b, c, d)				
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Paragraph for Question Nos. 1 to 2

There are four boxes B_1, B_2, B_3 and B_4 . Box B_i has i cards and on each card a number is printed, the numbers are from 1 to i . A box is selected randomly, the probability of selecting box B_i is $\frac{i}{10}$ and then a card is drawn.

Let E_i represent the event that a card with number 'i' is drawn. Then :

1. $P(E_1)$ is equal to :

(a) $\frac{1}{5}$

(b) $\frac{1}{10}$

(c) $\frac{2}{5}$

(d) $\frac{1}{4}$

2. $P(B_3|E_2)$ is equal to :

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) $\frac{2}{3}$

Paragraph for Question Nos. 3 to 5

Mr. A randomly picks 3 distinct numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and arranges them in descending order to form a three digit number. Mr. B randomly picks 3 distinct numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$ and also arranges them in descending order to form a 3 digit number.

- 3.** The probability that Mr. A's 3 digit number is always greater than Mr. B's 3 digit number is :

(a) $\frac{1}{9}$

(b) $\frac{1}{3}$

(c) $\frac{2}{3}$

(d) $\frac{1}{4}$

4. The probability that A and B has the same 3 digit number is :

(a) $\frac{7}{9}$

(b) $\frac{4}{9}$

(c) $\frac{1}{84}$

(d) $\frac{1}{72}$

5. The probability that Mr. A's number is larger than Mr. B's number, is :

(a) $\frac{37}{56}$

(b) $\frac{39}{56}$

(c) $\frac{31}{56}$

(d) none of these

Paragraph for Question Nos. 6 to 7

In an experiment a coin is tossed 10 times.

6. Probability that no two heads are consecutive is :

(a) $\frac{143}{2^{10}}$

(b) $\frac{9}{2^6}$

(c) $\frac{2^7 - 1}{2^{10}}$

(d) $\frac{2^6 - 1}{2^6}$

7. The probability of the event that "exactly four heads occur and occur alternately" is :

- (a) $1 - \frac{4}{2^{10}}$ (b) $1 - \frac{7}{2^{10}}$ (c) $\frac{4}{2^{10}}$ (d) $\frac{5}{2^{10}}$

Paragraph for Question Nos. 8 to 10

The rule of an "obstacle course" specifies that at the n^{th} obstacle a person has to toss a fair 6 sided die n times. If the sum of points in these n tosses is bigger than 2^n , the person is said to have crossed the obstacle.

8. The maximum obstacles a person can cross :

- (a) 4 (b) 5 (c) 6 (d) 7

9. The probability that a person crosses the first three obstacles :

- (a) $\frac{143}{216}$ (b) $\frac{100}{243}$ (c) $\frac{216}{243}$ (d) $\frac{100}{216}$

10. The probability that a person crosses the first two obstacles but fails to cross the third obstacle.

- (a) $\frac{36}{243}$ (b) $\frac{116}{216}$ (c) $\frac{35}{243}$ (d) $\frac{143}{243}$

Paragraph for Question Nos. 11 to 12

In an objective paper, there are two sections of 10 questions each. For 'section 1', each question has 5 options and only one option is correct and 'section 2' has 4 options with multiple answers and marks for a question in this section is awarded only if he ticks all correct answers. Marks for each question in 'section 1' is 1 and in 'section 2' is 3. (There is no negative marking).

11. If a candidate attempts only two questions by gussing, one from 'section 1' and one from 'section 2', the probability that he scores in both questions is :

- (a) $\frac{74}{75}$ (b) $\frac{1}{25}$ (c) $\frac{1}{15}$ (d) $\frac{1}{75}$

12. If a candidate in total attempts 4 questions all by gussing, then the probability of scoring 10 marks is :

- (a) $\frac{1}{15} \left(\frac{1}{15} \right)^2$ (b) $\frac{4}{5} \left(\frac{1}{15} \right)^3$ (c) $\frac{1}{5} \left(\frac{14}{15} \right)^3$ (d) None of these

Answers

1. (c)	2. (c)	3. (b)	4. (c)	5. (a)	6. (b)	7. (c)	8. (a)	9. (b)	10. (c)
11. (d)	12. (d)								

Exercise-4 : Matching Type Problems

1. A is a set containing n elements, A subset P (may be void also) is selected at random from set A and the set A is then reconstructed by replacing the elements of P . A subset Q (may be void also) of A is again chosen at random. The probability that

Column-I		Column-II	
(A)	Number of elements in P is equal to the number of elements in Q is	(P)	$\frac{{}^{2n}C_n}{4^n}$
(B)	The number of elements in P is more than that in Q is	(Q)	$\frac{(2^{2n} - {}^{2n}C_n)}{2^{2n+1}}$
(C)	$P \cap Q = \phi$ is	(R)	$\frac{{}^{2n}C_{n+1}}{4^n}$
(D)	Q is a subset of P is	(S)	$\left(\frac{3}{4}\right)^n$
		(T)	$\frac{{}^{2n}C_n}{4^{n-1}}$

Answers

1. $A \rightarrow P; B \rightarrow Q; C \rightarrow S; D \rightarrow S$

Exercise-5 : Subjective Type Problems

1. Mr. A writes an article. The article originally is error free. Each day Mr. B introduces one new error into the article. At the end of the day, Mr. A checks the article and has $\frac{2}{3}$ chance of catching each individual error still in the article. After 3 days, the probability that the article is error free can be expressed as $\frac{p}{q}$ where p and q are relatively prime positive integers. Let $\lambda = q - p$, then find the sum of the digits of λ .
2. India and Australia play a series of 7 one-day matches. Each team has equal probability of winning a match. No match ends in a draw. If the probability that India wins atleast three consecutive matches can be expressed as $\frac{p}{q}$ where p and q are relatively prime positive integers. Find the unit digit of p .
3. Two hunters A and B set out to hunt ducks. Each of them hits as often as he misses when shooting at ducks. Hunter A shoots at 50 ducks and hunter B shoots at 51 ducks. The probability that B bags more ducks than A can be expressed as $\frac{p}{q}$ in its lowest form. Find the value of $(p + q)$.
4. If $a, b, c \in N$, the probability that $a^2 + b^2 + c^2$ is divisible by 7 is $\frac{m}{n}$ where m, n are relatively prime natural numbers, then $m + n$ is equal to :
5. A fair coin is tossed 10 times. If the probability that heads never occur on consecutive tosses be $\frac{m}{n}$ (where m, n are coprime and $m, n \in N$), then the value of $(n - 7m)$ equals to :
6. A bag contains 2 red, 3 green and 4 black balls. 3 balls are drawn randomly and exactly 2 of them are found to be red. If p denotes the chance that one of the three balls drawn is green ; find the value of $7p$.
7. There are 3 different pairs (i.e., 6 units say a, a, b, b, c, c) of shoes in a lot. Now three person come and pick the shoes randomly (each gets 2 units). Let p be the probability that no one is able to wear shoes (i.e., no one gets a correct pair), then the value of $\frac{13p}{4-p}$, is :
8. A fair coin is tossed 12 times. If the probability that two heads do not occur consecutively is p , then the value of $\frac{[\sqrt{4096p-1}]}{2}$ is, where $[]$ denotes greatest integer function :
9. X and Y are two weak students in mathematics and their chances of solving a problem correctly are $\frac{1}{8}$ and $\frac{1}{12}$ respectively. They are given a question and they obtain the same answer. If the probability of common mistake is $\frac{1}{1001}$, then probability that the answer was correct is a/b (a and b are coprimes). Then $|a - b| =$

10. Seven digit numbers are formed using digits 1, 2, 3, 4, 5, 6, 7, 8, 9 without repetition. The probability of selecting a number such that product of any 5 consecutive digits is divisible by either 5 or 7 is P . Then $12P$ is equal to
11. Assume that for every person the probability that he has exactly one child, exactly 2 children and exactly 3 children are $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$ respectively. The probability that a person will have 4 grand children can be expressed as $\frac{p}{q}$ where p and q are relatively prime positive integers. Find the value of $5p - q$.
12. Mr. B has two fair 6-sided dice, one whose faces are numbered 1 to 6 and the second whose faces are numbered 3 to 8. Twice, he randomly picks one of dice (each dice equally likely) and rolls it. Given the sum of the resulting two rolls is 9. The probability he rolled same dice twice is $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $(m + n)$.

Answers

1.	7	2.	7	3.	3	4.	8	5.	1	6.	3	7.	2
8.	9	9.	1	10.	7	11.	7	12.	7				

□□□

Exercise-1 : Single Choice Problems

1. Solution set of the inequality $\log_{10^2} x - 3(\log_{10} x)(\log_{10}(x-2)) + 2\log_{10^2}(x-2) < 0$, is :
 (a) $(0, 4)$ (b) $(-\infty, 1)$ (c) $(4, \infty)$ (d) $(2, 4)$
2. The number of real solution/s of the equation $9^{\log_3(\log_e x)} = \log_e x - (\log_e x)^2 + 1$ is :
 (a) 0 (b) 1 (c) 2 (d) 3
3. If a, b, c are positive numbers such that $a^{\log_3 7} = 27$, $b^{\log_7 11} = 49$, $c^{\log_{11} 25} = \sqrt{11}$, then the sum of digits of $S = a^{(\log_3 7)^2} + b^{(\log_7 11)^2} + c^{(\log_{11} 25)^2}$ is :
 (a) 15 (b) 17 (c) 19 (d) 21
4. Least positive integral value of 'a' for which $\log_{\left(x+\frac{1}{x}\right)}(a^2 - 3a + 3) > 0$; $(x > 0)$:
 (a) 1 (b) 2 (c) 3 (d) 4
5. Let $P = \frac{5}{\frac{1}{\log_2 x} + \frac{1}{\log_3 x} + \frac{1}{\log_4 x} + \frac{1}{\log_5 x}}$ and $(120)^P = 32$, then the value of x be :
 (a) 1 (b) 2 (c) 3 (d) 4
6. If x, y, z be positive real numbers such that $\log_{2x}(z) = 3$, $\log_{5y}(z) = 6$ and $\log_{xy}(z) = \frac{2}{3}$ then the value of z is :
 (a) $\frac{1}{5}$ (b) $\frac{1}{10}$ (c) $\frac{3}{5}$ (d) $\frac{4}{9}$
7. Sum of values of x and y satisfying $\log_x(\log_3(\log_x y)) = 0$ and $\log_y 27 = 1$ is :
 (a) 27 (b) 30 (c) 33 (d) 36
8. $\log_{0.01} 1000 + \log_{0.1} 0.0001$ is equal to :
 (a) -2 (b) 3 (c) $-5/2$ (d) $5/2$

9. If $\log_{12} 27 = a$, then $\log_6 16 =$
 (a) $2\left(\frac{3-a}{3+a}\right)$ (b) $3\left(\frac{3-a}{3+a}\right)$ (c) $4\left(\frac{3-a}{3+a}\right)$ (d) None of these
10. If $\log_2(\log_2(\log_3 x)) = \log_2(\log_3(\log_2 y)) = 0$ then the value of $(x + y)$ is :
 (a) 17 (b) 9 (c) 21 (d) 19
11. Suppose that a and b are positive real numbers such that $\log_{27} a + \log_9 b = \frac{7}{2}$ and $\log_{27} b + \log_9 a = \frac{2}{3}$. Then the value of $a \cdot b$ is :
 (a) 81 (b) 243 (c) 27 (d) 729
12. If $2^a = 5$, $5^b = 8$, $8^c = 11$ and $11^d = 14$, then the value of 2^{abcd} is :
 (a) 1 (b) 2 (c) 7 (d) 14
13. Which of the following conditions necessarily imply that the real number x is rational ?
 (I) x^2 is rational (II) x^3 and x^5 are rational (III) x^2 and x^3 are rational
 (a) I and II only (b) I and III only (c) II and III only (d) III only
14. The value of $\frac{\log_8 17}{\log_9 23} - \frac{\log_{2\sqrt{2}} 17}{\log_3 23}$ is equal to :
 (a) -1 (b) 0 (c) $\frac{\log_2 17}{\log_3 23}$ (d) $\frac{4(\log_2 17)}{3(\log_3 23)}$
15. The true solution set of inequality $\log_{(2x-3)}(3x-4) > 0$ is equal to :
 (a) $\left(\frac{4}{3}, \frac{5}{3}\right) \cup (2, \infty)$ (b) $\left(\frac{3}{2}, \frac{5}{3}\right) \cup (2, \infty)$ (c) $\left(\frac{4}{3}, \frac{3}{2}\right) \cup (2, \infty)$ (d) $\left(\frac{2}{3}, \frac{4}{3}\right) \cup (2, \infty)$
16. If P is the number of natural numbers whose logarithm to the base 10 have the characteristic p and Q is the number of natural numbers logarithm of whose reciprocals to the base 10 have the characteristic $-q$ then $\log_{10} P - \log_{10} Q$ has the value equal to :
 (a) $p - q + 1$ (b) $p - q$ (c) $p + q - 1$ (d) $p - q - 1$
17. If $2^{2010} = a_n 10^n + a_{n-1} 10^{n-1} + \dots + a_2 10^2 + a_1 \cdot 10 + a_0$, where $a_i \in \{0, 1, 2, \dots, 9\}$ for all $i = 0, 1, 2, 3, \dots, n$, then $n =$
 (a) 603 (b) 604 (c) 605 (d) 606
18. The number of zeros after decimal before the start of any significant digit in the number $N = (0.15)^{20}$ are :
 (a) 15 (b) 16 (c) 17 (d) 18
19. $\log_2[\log_4(\log_{10} 16^4 + \log_{10} 25^8)]$ simplifies to :
 (a) an irrational (b) an odd prime
 (c) a composite (d) unity
20. The sum of all the solutions to the equation $2 \log x - \log(2x - 75) = 2$:
 (a) 30 (b) 350 (c) 75 (d) 200

21. $x^{\log_x a \cdot \log_a y \cdot \log_y z}$ is equal to :
 (a) x (b) y (c) z (d) x^z
22. Number of solution(s) of the equation $x^{x\sqrt{x}} = (x\sqrt{x})^x$ is/are :
 (a) 0 (b) 1 (c) 2 (d) 3
23. The difference of roots of the equation $(\log_{27} x^3)^2 = \log_{27} x^6$ is :
 (a) $\frac{2}{3}$ (b) 1 (c) 9 (d) 8
24. If $\log_{10} x + \log_{10} y = 2$, $x - y = 15$ then :
 (a) (x, y) lies on the line $y = 4x + 3$ (b) (x, y) lies on $y^2 = 4x$
 (c) (x, y) lies on $x = 4y$ (d) (x, y) lies on $4x = y$
25. Product of all values of x satisfying the equation
 $\sqrt{2^x \sqrt[3]{4^x (0.125)^{1/x}}} = 4(\sqrt[3]{2})$ is :
 (a) $\frac{14}{5}$ (b) 3 (c) $-\frac{1}{5}$ (d) $-\frac{3}{5}$
26. Sum of all values of x satisfying the equation
 $25^{(2x-x^2+1)} + 9^{(2x-x^2+1)} = 34(15^{(2x-x^2)})$ is :
 (a) 1 (b) 2 (c) 3 (d) 4
27. If $a^x = b^y = c^z = d^w$, then $\log_a(bcd) =$
 (a) $z\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{w}\right)$ (b) $y\left(\frac{1}{x} + \frac{1}{z} + \frac{1}{w}\right)$ (c) $x\left(\frac{1}{y} + \frac{1}{z} + \frac{1}{w}\right)$ (d) $\frac{xyz}{w}$
28. If $x = \frac{4}{(\sqrt{5}+1)(\sqrt[4]{5}+1)(\sqrt[8]{5}+1)(\sqrt[16]{5}+1)}$. Then the value of $(1+x)^{48}$ is :
 (a) 5 (b) 25 (c) 125 (d) 625
29. If $\log_x \log_{18}(\sqrt{2} + \sqrt{8}) = \frac{1}{3}$, then the value of $32x =$
 (a) 2 (b) 4 (c) 6 (d) 8
30. Let $n \in N$, $f(n) = \begin{cases} \log_8 n & \text{if } \log_8 n \text{ is integer} \\ 0 & \text{otherwise} \end{cases}$, then the value of $\sum_{n=1}^{2011} f(n)$ is :
 (a) 2011 (b) 2011×1006 (c) 6 (d) 2^{2011}
31. If the equation $\frac{\log_{12}(\log_8(\log_4 x))}{\log_5(\log_4(\log_y(\log_2 x)))} = 0$ has a solution for ' x ' when $c < y < b$, $y \neq a$, where ' b ' is as large as possible and ' c ' is as small as possible, then the value of $(a+b+c)$ is equals to :
 (a) 18 (b) 19 (c) 20 (d) 21

32. If $\log_{0.3}(x-1) < \log_{0.09}(x-1)$, then x lies in the interval :
 (a) $(2, \infty)$ (b) $(1, 2)$ (c) $(-2, -1)$ (d) $\left(1, \frac{3}{2}\right)$
33. The absolute integral value of the solution of the equation $\sqrt{7^{2x^2-5x-6}} = (\sqrt{2})^{3\log_2 49}$
 (a) 2 (b) 1 (c) 4 (d) 5
34. Let $1 \leq x \leq 256$ and M be the maximum value of $(\log_2 x)^4 + 16(\log_2 x)^2 \log_2 \left(\frac{16}{x}\right)$. The sum of the digits of M is :
 (a) 9 (b) 11 (c) 13 (d) 15
35. Let $1 \leq x \leq 256$ and M be the maximum value of $(\log_2 x)^4 + 16(\log_2 x)^2 \log_2 \left(\frac{16}{x}\right)$. The sum of the digits of M is :
 (a) 9 (b) 11 (c) 13 (d) 15
36. Number of real solution(s) of the equation $9\log_3(\log nx) = \ln x - (\ln^2 x) + 1$ is :
 (a) 0 (b) 1 (c) 2 (d) 3
37. The number of real values of the parameter λ for which $(\log_{16} x)^2 - \log_{16} x + \log_{16} \lambda = 0$ with real coefficients will have exactly one solution is :
 (a) 1 (b) 2 (c) 3 (d) 4
38. A rational number which is 50 times its own logarithm to the base 10 is :
 (a) 1 (b) 10 (c) 100 (d) 1000
39. If $x = \log_5(1000)$ and $y = \log_7(2058)$, then
 (a) $x > y$ (b) $x < y$ (c) $x = y$ (d) none of these
40. $7\log\left(\frac{16}{15}\right) + 5\log\left(\frac{25}{24}\right) + 3\log\left(\frac{81}{80}\right)$ is equal to :
 (a) 0 (b) 1 (c) $\log 2$ (d) $\log 3$
41. $\log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \dots + \log_{10} \tan 89^\circ$ is equal to :
 (a) 0 (b) 1 (c) 27 (d) 81
42. $\log_7 \log_7 \sqrt{7\sqrt{7\sqrt{7}}}$ is equal to :
 (a) $3\log_2 7$ (b) $3\log_7 2$ (c) $1 - 3\log_7 2$ (d) $1 - 3\log_2 7$
43. If $(4)^{\log_9 3} + (9)^{\log_2 4} = (10)^{\log_x 83}$, then x is equal to :
 (a) 2 (b) 3 (c) 10 (d) 30
44. $x^{\log_{10}\left(\frac{y}{z}\right)} \cdot y^{\log_{10}\left(\frac{z}{x}\right)} \cdot z^{\log_{10}\left(\frac{x}{y}\right)}$ is equal to :
 (a) 0 (b) 1 (c) -1 (d) 2

45. The solution set of the equation : $\log_x 2 \log_{2x} 2 = \log_{4x} 2$ is :
 (a) $\{2^{-\sqrt{2}}, 2^{\sqrt{2}}\}$ (b) $\{1/2, 2\}$ (c) $\{1/4, 2^2\}$ (d) none of these
46. The least value of the expression $2 \log_{10} x - \log_x 0.01$ is ($x > 1$)
 (a) 2 (b) 4 (c) 6 (d) 8
47. If $\sqrt{\log_2 x} - 0.5 = \log_2 \sqrt{x}$, then x equals to :
 (a) odd integer (b) prime number
 (c) composite number (d) irrational
48. If x_1 and x_2 are the roots of the equation $e^{2x} \ln x = x^3$ with $x_1 > x_2$, then
 (a) $x_1 = 2x_2$ (b) $x_1 = x_2^2$ (c) $2x_1 = x_2^2$ (d) $x_1^2 = x_2^2$
49. Let M denote $\text{antilog}_{32} 0.6$ and N denote the value of $49^{(1-\log_7 2)} + 5^{-\log_5 4}$. Then $M.N$ is :
 (a) 100 (b) 400 (c) 50 (d) 200
50. If $\log_2(\log_2(\log_3 x)) = \log_3(\log_3(\log_2 y)) = 0$, then $x - y$ is equal to :
 (a) 0 (b) 1 (c) 8 (d) 9
51. $\left| \log_{\frac{1}{2}} 10 + \left| \log_4 625 - \left| \log_{\frac{1}{2}} 5 \right| \right| \right| =$
 (a) $\log_{1/2} 2$ (b) $\log_2 5$ (c) $\log_2 2$ (d) $\log_2 25$
52. If $\log_4 5 = a$ and $\log_5 6 = b$, then $\log_3 2$ is equal to :
 (a) $\frac{1}{2a+1}$ (b) $\frac{1}{2b+1}$ (c) $2ab+1$ (d) $\frac{1}{2ab-1}$
53. If $x = \log_a bc$; $y = \log_b ac$ and $z = \log_c ab$ then which of the following is equal to unity ?
 (a) $x+y+z$ (b) xyz
 (c) $\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z}$ (d) $(1+x) + (1+y) + (1+z)$
54. $x^{\log_x a \cdot \log_a y \cdot \log_y z}$ is equal to :
 (a) x (b) y (c) z (d) a
55. Number of value(s) of ' x ' satisfying the equation $x^{\log_{\sqrt{x}}(x-3)} = 9$ is/are
 (a) 0 (b) 1 (c) 2 (d) 6
56. $\log_{0.01} 1000 + \log_{0.01} 0.0001$ is equal to :
 (a) -2 (b) 3 (c) $-\frac{5}{2}$ (d) $\frac{5}{2}$
57. If $7 \log_a \frac{16}{15} + 5 \log_a \frac{25}{24} + 3 \log_a \frac{81}{80} = 8$, then $a =$
 (a) $2^{1/8}$ (b) $(10)^{1/8}$ (c) $(30)^{1/8}$ (d) 1

58. $\log_8(128) - \log_9 \cot\left(\frac{\pi}{3}\right) =$
 (a) $\frac{31}{12}$ (b) $\frac{19}{12}$ (c) $\frac{13}{12}$ (d) $\frac{11}{12}$
59. The value of $\left(\frac{1}{\sqrt{27}}\right)^{2 - \left(\frac{\log_5 16}{2 \log_5 9}\right)}$ equals to :
 (a) $\frac{5\sqrt{2}}{27}$ (b) $\frac{\sqrt{2}}{27}$ (c) $\frac{4\sqrt{2}}{27}$ (d) $\frac{2\sqrt{2}}{27}$
60. The sum of all the roots of the equation $\log_2(x-1) + \log_2(x+2) - \log_2(3x-1) = \log_2 4$
 (a) 12 (b) 2 (c) 10 (d) 11
61. $\frac{(\log_{100} 10)(\log_2(\log_4 2))(\log_4 \log_2^2(256)^2)}{\log_4 8 + \log_8 4} =$
 (a) $-\frac{6}{13}$ (b) $-\frac{1}{2}$ (c) $-\frac{8}{13}$ (d) $-\frac{12}{13}$
62. Let $\lambda = \log_5 \log_5(3)$. If $3^{k+5^{-\lambda}} = 405$, then the value of k is :
 (a) 3 (b) 5 (c) 4 (d) 6
63. A circle has a radius $\log_{10}(a^2)$ and a circumference of $\log_{10}(b^4)$. Then the value of $\log_a b$ is equal to :
 (a) $\frac{1}{4\pi}$ (b) $\frac{1}{\pi}$ (c) 2π (d) π
64. If $2^x = 3^y = 6^{-z}$, the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ is equal to :
 (a) 0 (b) 1 (c) 2 (d) 3
65. The value of $\log_{(\sqrt{2}-1)}(5\sqrt{2}-7)$ is :
 (a) 0 (b) 1 (c) 2 (d) 3
66. The value of $\log_{ab}\left(\frac{\sqrt[3]{a}}{\sqrt{b}}\right)$, if $\log_{ab} a = 4$ is equal to :
 (a) 2 (b) $\frac{13}{6}$ (c) $\frac{15}{6}$ (d) $\frac{17}{6}$
67. Identify the correct option
 (a) $\log_2 3 < \log_{1/4} 5$ (b) $\log_5 7 < \log_8 3$
 (c) $\log_{\sqrt[3]{2}} \sqrt{3} > \log_{\sqrt[3]{2}} \sqrt{5}$ (d) $2^{\frac{1}{4}} > \left(\frac{3}{2}\right)^{1/3}$
68. Sum of all values of x satisfying the system of equations $5(\log_y x + \log_x y) = 26$, $xy = 64$ is :
 (a) 42 (b) 34 (c) 32 (d) 2

69. The product of all values of x satisfying the equations $\log_3 a - \log_x a = \log_{x/3} a$ is :

- (a) 3 (b) $\frac{3}{2}$ (c) 18 (d) 27

70. The value of $x + y + z$ satisfying the system of equations

$$\log_2 x + \log_4 y + \log_4 z = 2 \text{ is}$$

$$\log_3 y + \log_9 z + \log_9 x = 2$$

$$\log_4 z + \log_{16} x + \log_{16} y = 2$$

- (a) $\frac{175}{12}$ (b) $\frac{349}{24}$ (c) $\frac{353}{24}$ (d) $\frac{112}{3}$

71. $\left(\frac{1}{49}\right)^{1+\log_7 2} + 5^{-\log_5 7} =$

- (a) $7\frac{1}{196}$ (b) $7\frac{3}{196}$ (c) $7\frac{5}{196}$ (d) $7\frac{1}{98}$

72. The number of real values of x satisfying the equation $\log_2(3-x) - \log_2 \left(\frac{\sin\left(\frac{3\pi}{4}\right)}{(5-x)} \right) = \frac{1}{2} + \log_2(x+7)$ is :

- (a) 0 (b) 1 (c) 2 (d) 3

73. If $\log_k x \log_5 k = \log_x 5$, $k \neq 1$, $k > 0$, then sum of all values of x is :

- (a) 5 (b) $\frac{24}{5}$ (c) $\frac{26}{5}$ (d) $\frac{37}{5}$

74. The product of all values of x satisfying the equation $|x-1|^{\log_3 x^2 - 2\log_x 9} = (x-1)^7$, is :

- (a) 162 (b) $\frac{162}{\sqrt{3}}$ (c) $\frac{81}{\sqrt{3}}$ (d) 81

75. The number of values of x satisfying the equation $\log_2(9^{x-1} + 7) = 2 + \log_2(3^{x-1} + 1)$ is :

- (a) 1 (b) 2 (c) 3 (d) 0

76. Which is the correct order for a given number α , $\alpha > 1$

- (a) $\log_2 \alpha < \log_3 \alpha < \log_e \alpha < \log_{10} \alpha$ (b) $\log_{10} \alpha < \log_3 \alpha < \log_e \alpha < \log_2 \alpha$
 (c) $\log_{10} \alpha < \log_e \alpha < \log_2 \alpha < \log_3 \alpha$ (d) $\log_3 \alpha < \log_e \alpha < \log_2 \alpha < \log_{10} \alpha$

77. Let $1 \leq x \leq 256$ and M be the maximum value of $(\log_2 x)^4 + 16(\log_2 x)^2 \log_2 \left(\frac{16}{x}\right)$. The sum of the digits of M is :

- (a) 9 (b) 11 (c) 13 (d) 15

Exercise-2 : One or More than One Answer is/are Correct

1. The values of 'x' satisfies the equation $\frac{1 - 2(\log x^2)^2}{\log x - 2(\log x)^2} = 1$ (is/are) :

(where log is logarithm to the base 10)

- (a) $\frac{1}{\sqrt{10}}$ (b) $\frac{1}{\sqrt{20}}$ (c) $\sqrt[3]{10}$ (d) $\sqrt{10}$

2. If $\log_a x = b$ for permissible values of a and x then identify the statement(s) which can be correct?

- (a) If a and b are two irrational numbers then x can be rational.
 (b) If a rational and b irrational then x can be rational.
 (c) If a irrational and b rational then x can be rational.
 (d) If a rational and b rational then x can be rational.

3. Consider the quadratic equation, $(\log_{10} 8)x^2 - (\log_{10} 5)x = 2(\log_2 10)^{-1} - x$. Which of the following quantities are irrational ?

- (a) Sum of the roots (b) Product of the roots
 (c) Sum of the coefficients (d) Discriminant

4. Let $A = \text{Minimum}(x^2 - 2x + 7)$, $x \in R$ and $B = \text{Minimum}(x^2 - 2x + 7)$, $x \in [2, \infty)$, then :

- (a) $\log_{(B-A)}(A+B)$ is not defined (b) $A+B=13$
 (c) $\log_{(2B-A)} A < 1$ (d) $\log_{(2A-B)} A > 1$

Answers

1.	(a, c)	2.	(a, b, c, d)	3.	(c, d)	4.	(a, b, c, d)				
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Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 3

Let $\log_3 N = \alpha_1 + \beta_1$

$\log_5 N = \alpha_2 + \beta_2$

$\log_7 N = \alpha_3 + \beta_3$

where α_1, α_2 and α_3 are integers and $\beta_1, \beta_2, \beta_3 \in [0, 1)$.

- Number of integral values of N if $\alpha_1 = 4$ and $\alpha_2 = 2$:
(a) 46 (b) 45 (c) 44 (d) 47
- Largest integral value of N if $\alpha_1 = 5, \alpha_2 = 3$ and $\alpha_3 = 2$.
(a) 342 (b) 343 (c) 243 (d) 242
- Difference of largest and smallest integral values of N if $\alpha_1 = 5, \alpha_2 = 3$ and $\alpha_3 = 2$.
(a) 97 (b) 100 (c) 98 (d) 99

Paragraph for Question Nos. 4 to 5

If $\log_{10}|x^3 + y^3| - \log_{10}|x^2 - xy + y^2| + \log_{10}|x^3 - y^3| - \log_{10}|x^2 + xy + y^2| = \log_{10} 221$.

Where x, y are integers, then

- If $x = 111$, then y can be :
(a) ± 111 (b) ± 2 (c) ± 110 (d) ± 109
- If $y = 2$, then value of x can be :
(a) ± 111 (b) ± 15 (c) ± 2 (d) ± 110

Paragraph for Question Nos. 6 to 7

Given a right triangle ABC right angled at C and whose legs are given $1 + 4\log_{p^2}(2p)$, $1 + 2^{\log_2(\log_2 p)}$ and hypotenuse is given to be $1 + \log_2(4p)$. The area of $\triangle ABC$ and circle circumscribing it are Δ_1 and Δ_2 respectively, then

- $\Delta_1 + \frac{4\Delta_2}{\pi}$ is equal to :
(a) 31 (b) 28 (c) $3 + \frac{1}{\sqrt{2}}$ (d) 199
- The value of $\sin\left(\frac{\pi(25p^2\Delta_1 + 2)}{6}\right) =$
(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{\sqrt{3}}{2}$ (d) 1

Answers

1.	(c)	2.	(a)	3.	(d)	4.	(c)	5.	(b)	6.	(a)	7.	(c)						
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Exercise-4 : Matching Type Problems

1.

Column-I		Column-II	
(A)	If $a = 3(\sqrt{8+2\sqrt{7}} - \sqrt{8-2\sqrt{7}})$, $b = \sqrt{(42)(30) + 36}$, then the value of $\log_a b$ is equal to	(P)	-1
(B)	If $a = (\sqrt{4+2\sqrt{3}} - \sqrt{4-2\sqrt{3}})$, $b = \sqrt{11+6\sqrt{2}} - \sqrt{11-6\sqrt{2}}$ then the value of $\log_a b$ is equal to	(Q)	1
(C)	If $a = \sqrt{3+2\sqrt{2}}$, $b = \sqrt{3-2\sqrt{2}}$, then the value of $\log_a b$ is equal to	(R)	2
(D)	If $a = \sqrt{7+\sqrt{7^2-1}}$, $b = \sqrt{7-\sqrt{7^2-1}}$, then the value of $\log_a b$ is equal to	(S)	$\frac{3}{2}$
		(T)	None of these

Answers

1. $A \rightarrow R; B \rightarrow S; C \rightarrow P; D \rightarrow P$

Exercise-5 : Subjective Type Problems

- The number $N = 6^{\log_{10} 40} \cdot 5^{\log_{10} 36}$ is a natural number. Then sum of digits of N is :
- The minimum value of 'c' such that $\log_b(a^{\log_2 b}) = \log_a(b^{\log_2 b})$ and $\log_a(c - (b - a)^2) = 3$, where $a, b \in N$ is :
- How many positive integers b have the property that $\log_b 729$ is a positive integer ?
- The number of negative integral values of x satisfying the inequality $\log_{\left(x+\frac{5}{2}\right)}\left(\frac{x-5}{2x-3}\right)^2 < 0$ is :
- $\frac{6}{5} a^{(\log_a x)(\log_{10} a)(\log_a 5)} - 3^{\log_{10}\left(\frac{x}{10}\right)} = 9^{\log_{100} x + \log_4 2}$ (where $a > 0, a \neq 1$), then $\log_3 x = \alpha + \beta$, α is integer, $\beta \in [0, 1)$, then $\alpha =$
- If $\log_5\left(\frac{a+b}{3}\right) = \frac{\log_5 a + \log_5 b}{2}$, then $\frac{a^4 + b^4}{a^2 b^2} =$
- Let a, b, c, d are positive integers such that $\log_a b = \frac{3}{2}$ and $\log_c d = \frac{5}{4}$. If $(a - c) = 9$. Find the value of $(b - d)$.
- The number of real values of x satisfying the equation $\log_{10} \sqrt{1+x} + 3 \log_{10} \sqrt{1-x} = 2 + \log_{10} \sqrt{1-x^2}$ is :
- The ordered pair (x, y) satisfying the equation $x^2 = 1 + 6 \log_4 y$ and $y^2 = 2^x y + 2^{2x+1}$ are (x_1, y_1) and (x_2, y_2) , then find the value of $\log_2 |x_1 x_2 y_1 y_2|$.
- If $\log_7 \log_7 \sqrt{7\sqrt{7\sqrt{7}}} = 1 - a \log_7 2$ and $\log_{15} \log_{15} \sqrt{15\sqrt{15\sqrt{15\sqrt{15}}}} = 1 - b \log_{15} 2$, then $a + b =$
- The number of ordered pair(s) of (x, y) satisfying the equations $\log_{(1+x)}(1 - 2y + y^2) + \log_{(1-y)}(1 + 2x + x^2) = 4$ and $\log_{(1+x)}(1 + 2y) + \log_{(1-y)}(1 + 2x) = 2$
- If $\log_b n = 2$ and $\log_n(2b) = 2$, then $nb =$
- If $\log_y x + \log_x y = 2$, and $x^2 + y = 12$, then the value of xy is :
- If x, y satisfy the equation, $y^x = x^y$ and $x = 2y$, then $x^2 + y^2 =$
- Find the number of real values of x satisfying the equation.

$$\log_2(4^{x+1} + 4) \cdot \log_2(4^x + 1) = \log_{1/\sqrt{2}} \sqrt{\frac{1}{8}}$$
- If $x_1, x_2 (x_1 > x_2)$ are the two solutions of the equation $3^{\log_2 x} - 12(x^{\log_{16} 9}) = \log_3 \left(\frac{1}{3}\right)^{3^3}$, then the value of $x_1 - 2x_2$ is :

17. Find the number of real values of x satisfying the equation $9^{2\log_9 x} + 4x + 3 = 0$.

18. If $\log_{16}(\log_{\sqrt[3]{3}}(\log_{\sqrt[5]{5}}(x))) = \frac{1}{2}$; find x .

19. The value $\left[\frac{1}{6} \left(\frac{2\log_{10}(1728)}{1 + \frac{1}{2}\log_{10}(0.36) + \frac{1}{3}\log_{10} 8} \right)^{1/2} \right]^{-1}$ is :

Answers

1.	9	2.	8	3.	4	4.	0	5.	4	6.	47	7.	93
8.	0	9.	7	10.	7	11.	1	12.	2	13.	9	14.	20
15.	1	16.	8	17.	0	18.	5	19.	2				

Co-ordinate Geometry

17. Straight Lines

18. Circle

19. Parabola

20. Ellipse

21. Hyperbola

Chapter 17 – Straight Lines

Exercise-1 : Single Choice Problems

- The ratio in which the line segment joining $(2, -3)$ and $(5, 6)$ is divided by the x -axis is :
 (a) $3 : 1$ (b) $1 : 2$
 (c) $\sqrt{3} : 2$ (d) $\sqrt{2} : 3$
- If L is the line whose equation is $ax + by = c$. Let M be the reflection of L through the y -axis, and let N be the reflection of L through the x -axis. Which of the following must be true about M and N for all choices of a , b and c ?
 (a) The x -intercepts of M and N are equal
 (b) The y -intercepts of M and N are equal
 (c) The slopes of M and N are equal
 (d) The slopes of M and N are reciprocal
- The complete set of real values of ' a ' such that the point $P(a, \sin a)$ lies inside the triangle formed by the lines $x - 2y + 2 = 0$; $x + y = 0$ and $x - y - \pi = 0$, is :
 (a) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ (b) $\left(\frac{\pi}{2}, \pi\right) \cup \left(\frac{2\pi}{2}, 2\pi\right)$
 (c) $(0, \pi)$ (d) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$
- Let m be a positive integer and let the lines $13x + 11y = 700$ and $y = mx - 1$ intersect in a point whose coordinates are integer. Then m equals to :
 (a) 4 (b) 5 (c) 6 (d) 7
- If $P = \left(\frac{1}{x_p}, p\right)$; $Q = \left(\frac{1}{x_q}, q\right)$; $R = \left(\frac{1}{x_r}, r\right)$
 where $x_k \neq 0$, denotes the k^{th} terms of a H.P. for $k \in N$, then:

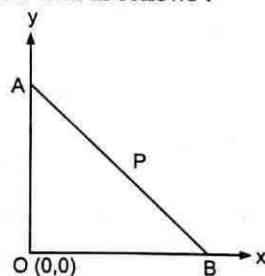
- (a) $\text{ar. } (\Delta PQR) = \frac{p^2 q^2 r^2}{2} \sqrt{(p-q)^2 + (q-r)^2 + (r-p)^2}$
- (b) ΔPQR is a right angled triangle
- (c) the points P, Q, R are collinear
- (d) None of these
6. If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then c has the value :
- (a) 1 (b) -1 (c) 2 (d) -2
7. A piece of cheese is located at $(12, 10)$ in a coordinate plane. A mouse is at $(4, -2)$ and is running up the line $y = -5x + 18$. At the point (a, b) , the mouse starts getting farther from the cheese rather than closer to it. The value of $(a + b)$ is:
- (a) 6 (b) 10
- (c) 18 (d) 14
8. The vertex of right angle of a right angled triangle lies on the straight line $2x + y - 10 = 0$ and the two other vertices, at points $(2, -3)$ and $(4, 1)$ then the area of triangle in sq. units is:
- (a) $\sqrt{10}$ (b) 3 (c) $\frac{33}{5}$ (d) 11
9. Given the family of lines, $a(2x + y + 4) + b(x - 2y - 3) = 0$. Among the lines of the family, the number of lines situated at a distance of $\sqrt{10}$ from the point $M(2, -3)$ is:
- (a) 0 (b) 1
- (c) 2 (d) ∞
10. Point $(0, \beta)$ lies on or inside the triangle formed by the lines $y = 0$, $x + y = 8$ and $3x - 4y + 12 = 0$. Then β can be :
- (a) 2 (b) 4 (c) 8 (d) 12
11. If the lines $x + y + 1 = 0$, $4x + 3y + 4 = 0$ and $x + \alpha y + \beta = 0$, where $\alpha^2 + \beta^2 = 2$, are concurrent then:
- (a) $\alpha = 1, \beta = -1$ (b) $\alpha = 1, \beta = \pm 1$
- (c) $\alpha = -1, \beta = \pm 1$ (d) $\alpha = \pm 1, \beta = 1$
12. A straight line through the origin 'O' meets the parallel lines $4x + 2y = 9$ and $2x + y = -6$ at points P and Q respectively. Then the point 'O' divides the segment PQ in the ratio :
- (a) 1 : 2 (b) 4 : 3 (c) 2 : 1 (d) 3 : 4
13. If the points $(2a, a)$, $(a, 2a)$ and (a, a) enclose a triangle of area 72 units, then co-ordinates of the centroid of the triangle may be :
- (a) $(4, 4)$ (b) $(-4, 4)$ (c) $(12, 12)$ (d) $(16, 16)$
14. Let $g(x) = ax + b$, where $a < 0$ and g is defined from $[1, 3]$ onto $[0, 2]$ then the value of $\cot(\cos^{-1}(|\sin x| + |\cos x|) + \sin^{-1}(-|\cos x| - |\sin x|))$ is equal to :
- (a) $g(1)$ (b) $g(2)$ (c) $g(3)$ (d) $g(1) + g(3)$

15. If the distances of any point P from the points $A(a+b, a-b)$ and $B(a-b, a+b)$ are equal, then locus of P is :
(a) $ax+by=0$ (b) $ax-by=0$ (c) $bx+ay=0$ (d) $x-y=0$
16. If the equation $4y^3 - 8a^2yx^2 - 3ay^2x + 8x^3 = 0$ represent three straight lines, two of them are perpendicular then sum of all possible values of a is equal to :
(a) $\frac{3}{8}$ (b) $\frac{-3}{4}$ (c) $\frac{1}{4}$ (d) -2
17. The orthocentre of the triangle formed by the lines $x-7y+6=0$, $2x-5y-6=0$ and $7x+y-8=0$ is :
(a) $(8, 2)$ (b) $(0, 0)$ (c) $(1, 1)$ (d) $(2, 8)$
18. All the chords of the curve $2x^2 + 3y^2 - 5x = 0$ which subtend a right angle at the origin are concurrent at :
(a) $(0, 1)$ (b) $(1, 0)$ (c) $(-1, 1)$ (d) $(1, -1)$
19. From a point $P \equiv (3, 4)$ perpendiculars PQ and PR are drawn to line $3x+4y-7=0$ and a variable line $y-1=m(x-7)$ respectively, then maximum area of ΔPQR is :
(a) 10 (b) 12 (c) 6 (d) 9
20. The equation of two adjacent sides of rhombus are given by $y=x$ and $y=7x$. The diagonals of the rhombus intersect each other at the point $(1, 2)$. Then the area of the rhombus is :
(a) $\frac{10}{3}$ (b) $\frac{20}{3}$ (c) $\frac{40}{3}$ (d) $\frac{50}{3}$
21. The point $P(3, 3)$ is reflected across the line $y=-x$. Then it is translated horizontally 3 units to the left and vertically 3 units up. Finally, it is reflected across the line $y=x$. What are the coordinates of the point after these transformations ?
(a) $(0, -6)$ (b) $(0, 0)$
(c) $(-6, 6)$ (d) $(-6, 0)$
22. The equations $x=t^3+9$ and $y=\frac{3t^3}{4}+6$ represents a straight line where t is a parameter. Then y -intercept of the line is :
(a) $-\frac{3}{4}$ (b) 9 (c) 6 (d) 1
23. The combined equation of two adjacent sides of a rhombus formed in first quadrant is $7x^2 - 8xy + y^2 = 0$; then slope of its longer diagonal is :
(a) $-\frac{1}{2}$ (b) -2 (c) 2 (d) $\frac{1}{2}$
24. The number of integral points inside the triangle made by the line $3x+4y-12=0$ with the coordinate axes which are equidistant from at least two sides is/are :
(an integral point is a point both of whose coordinates are integers.)
(a) 1 (b) 2 (c) 3 (d) 4

25. The area of triangle formed by the straight lines whose equations are $y = 4x + 2$, $2y = x + 3$ and $x = 0$ is :
- (a) $\frac{25}{7\sqrt{2}}$ (b) $\frac{\sqrt{2}}{28}$ (c) $\frac{1}{28}$ (d) $\frac{15}{7}$
26. In a triangle ABC , if A is $(1, 2)$ and the equations of the medians through B and C are $x + y = 5$ and $x = 4$ respectively then B must be :
- (a) $(1, 4)$ (b) $(7, -2)$ (c) $(4, 1)$ (d) $(-2, 7)$
27. The equation of image of pair of lines $y = |x - 1|$ with respect to y -axis is :
- (a) $x^2 - y^2 - 2x + 1 = 0$ (b) $x^2 - y^2 - 4x + 4 = 0$
 (c) $4x^2 - 4x - y^2 + 1 = 0$ (d) $x^2 - y^2 + 2x + 1 = 0$
28. If P , Q and R are three points with coordinates $(1, 4)$, $(4, 5)$ and (m, m) respectively, then the value of m for which $PR + RQ$ is minimum, is :
- (a) 4 (b) 3 (c) $\frac{17}{8}$ (d) $\frac{7}{2}$
29. The vertices of triangle ABC are $A(-1, -7)$, $B(5, 1)$ and $C(1, 4)$. The equation of the bisector of the angle ABC of $\triangle ABC$ is :
- (a) $y + 2x - 11 = 0$ (b) $x - 7y + 2 = 0$
 (c) $y - 2x + 9 = 0$ (d) $y + 7x - 36 = 0$
30. If one of the lines given by $6x^2 - xy + 4cy^2 = 0$ is $3x + 4y = 0$, then $c =$
- (a) -3 (b) -1 (c) 3 (d) 1
31. The equations of L_1 and L_2 are $y = mx$ and $y = nx$, respectively. Suppose L_1 make twice as large of an angle with the horizontal (measured counterclockwise from the positive x -axis) as does L_2 and that L_1 has 4 times the slope of L_2 . If L_1 is not horizontal, then the value of the product (mn) equals:
- (a) $\frac{\sqrt{2}}{2}$ (b) $-\frac{\sqrt{2}}{2}$
 (c) 2 (d) -2
32. Given $A(0, 0)$ and $B(x, y)$ with $x \in (0, 1)$ and $y > 0$. Let the slope of the line AB equals m_1 . Point C lies on the line $x = 1$ such that the slope of BC equals m_2 where $0 < m_2 < m_1$. If the area of the triangle ABC can be expressed as $(m_1 - m_2)f(x)$, then the largest possible value of $f(x)$ is:
- (a) 1 (b) $1/2$
 (c) $1/4$ (d) $1/8$
33. If non-zero numbers a, b, c are in H.P., then the straight line $\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0$ always passes through a fixed point, co-ordinate of fixed point is :
- (a) $(-1, 2)$ (b) $(-1, -2)$ (c) $(1, -2)$ (d) $\left(1, \frac{1}{2}\right)$

34. If $\frac{x^2}{a} + \frac{y^2}{b} + \frac{2xy}{h} = 0$ represent pair of straight lines and slope of one line is twice the other, then $ab : h^2$ is :
- (a) 9 : 8 (b) 8 : 9 (c) 1 : 2 (d) 2 : 1
35. **Statement-1:** A variable line drawn through a fixed point cuts the coordinate axes at A and B. The locus of mid-point of AB is a circle.
because
Statement-2: Through 3 non-collinear points in a plane, only one circle can be drawn.
- (a) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(b) Statement-1 is true, statement-2 is true and statement-2 is not the correct explanation for statement-1.
(c) Statement-1 is true, statement-2 is false.
(d) Statement-1 is false, statement-2 is true.
36. A line passing through origin and is perpendicular to two parallel lines $2x + y + 6 = 0$ and $4x + 2y - 9 = 0$, then the ratio in which the origin divides this line segment is :
- (a) 1 : 2 (b) 1 : 1
(c) 5 : 4 (d) 3 : 4
37. If a vertex of a triangle is (1, 1) and the mid-points of two sides through this vertex are (-1, 2) and (3, 2), then the centroid of the triangle is :
- (a) $\left(-1, \frac{7}{3}\right)$ (b) $\left(-\frac{1}{3}, \frac{7}{3}\right)$ (c) $\left(1, \frac{7}{3}\right)$ (d) $\left(\frac{1}{3}, \frac{7}{3}\right)$
38. The diagonals of parallelogram PQRS are along the lines $x + 3y = 4$ and $6x - 2y = 7$. Then PQRS must be :
- (a) rectangle (b) square
(c) rhombus (d) neither rhombus nor rectangle
39. The two points on the line $x + y = 4$ that lie at a unit perpendicular distance from the line $4x + 3y = 10$ are (a_1, b_1) and (a_2, b_2) , then $a_1 + b_1 + a_2 + b_2 =$
- (a) 5 (b) 6 (c) 7 (d) 8
40. The orthocentre of the triangle formed by the lines $x + y = 1$, $2x + 3y = 6$ and $4x - y + 4 = 0$ lies in :
- (a) first quadrant (b) second quadrant
(c) third quadrant (d) fourth quadrant
41. The equation of the line passing through the intersection of the lines $3x + 4y = -5$, $4x + 6y = 6$ and perpendicular to $7x - 5y + 3 = 0$ is :
- (a) $5x + 7y - 2 = 0$ (b) $5x - 7y + 2 = 0$
(c) $7x - 5y + 2 = 0$ (d) $5x + 7y + 2 = 0$

42. The points (2, 1), (8, 5) and (x, 7) lie on a straight line. Then the value of x is :
 (a) 10 (b) 11 (c) 12 (d) $\frac{35}{3}$
43. In a parallelogram PQRS (taken in order), P is the point (-1, -1), Q is (8, 0) and R is (7, 5). Then S is the point :
 (a) (-1, 4) (b) (-2, 2) (c) $\left(-2, \frac{7}{2}\right)$ (d) (-2, 4)
44. The area of triangle whose vertices are (a, a), (a + 1, a + 1), (a + 2, a) is :
 (a) a^3 (b) 2a (c) 1 (d) 2
45. The equation $x^2 + y^2 - 2xy - 1 = 0$ represents :
 (a) two parallel straight lines (b) two perpendicular straight lines
 (c) a point (d) a circle
46. Let A $\equiv$ (-2, 0) and B $\equiv$ (2, 0), then the number of integral values of a, $a \in [-10, 10]$ for which line segment AB subtends an acute angle at point C $\equiv$ (a, a + 1) is :
 (a) 15 (b) 17 (c) 19 (d) 21
47. The angle between sides of a rhombus whose $\sqrt{2}$ times sides is mean of its two diagonal, is equal to :
 (a) 300° (b) 45° (c) 60° (d) 90°
48. A rod of AB of length 3 rests on a wall as follows :



P is a point on AB such that $AP : PB = 1 : 2$. If the rod slides along the wall, then the locus of P lies on

- (a) $2x + y + xy = 2$ (b) $4x^2 + xy + xy + y^2 = 4$
 (c) $4x^2 + y^2 = 4$ (d) $x^2 + y^2 - x - 2y = 0$
49. If $\frac{x^2}{a} + \frac{y^2}{b} + \frac{2xy}{h} = 0$, represents pair of straight lines and slope of one line is twice the other. Then $ab : h^2$ is :
 (a) 8 : 9 (b) 1 : 2 (c) 2 : 1 (d) 9 : 8

50. Locus of point of reflection of point $(a, 0)$ w.r.t. the line $yt = x + at^2$ is given by (t is parameter, $t \in \mathbb{R}$):
 (a) $x - a = 0$ (b) $y - a = 0$ (c) $x + a = 0$ (d) $y + a = 0$
51. A light ray emerging from the point source placed at $P(1, 3)$ is reflected at a point Q in the x -axis. If the reflected ray passes through $R(6, 7)$, then abscissa of Q is :
 (a) $\frac{5}{2}$ (b) 3 (c) $\frac{7}{2}$ (d) 1
52. If the axes are rotated through 60° in the anticlockwise sense, find the transformed form of the equation $x^2 - y^2 = a^2$:
 (a) $X^2 + Y^2 - 3\sqrt{3}XY = 2a^2$ (b) $X^2 + Y^2 = a^2$
 (c) $Y^2 - X^2 - 2\sqrt{3}XY = 2a^2$ (d) $X^2 - Y^2 + 2\sqrt{3}XY = 2a^2$
53. The straight line $3x + y - 4 = 0$, $x + 3y - 4 = 0$ and $x + y = 0$ form a triangle which is :
 (a) equilateral (b) right-angled
 (c) acute-angled and isosceles (d) obtuse-angled and isosceles
54. If m and b are real numbers and $mb > 0$, then the line whose equation is $y = mx + b$ cannot contain the point:
 (a) $(0, 2008)$ (b) $(2008, 0)$
 (c) $(0, -2008)$ (d) $(20, -100)$
55. The number of possible straight lines, passing through $(2, 3)$ and forming a triangle with coordinate axes, whose area is 12 sq. units, is:
 (a) one (b) two
 (c) three (d) four
56. If x_1, x_2, x_3 and y_1, y_2, y_3 are both in G.P with the same common ratio then the points (x_1, y_1) , (x_2, y_2) and (x_3, y_3)
 (a) lie on a straight line (b) lie on a circle
 (c) are vertices of a triangle (d) None of these
57. Locus of centroid of the triangle whose vertices are $(a \cos t, a \sin t)$, $(b \sin t, -b \cos t)$ and $(1, 0)$; where t is a parameter is :
 (a) $(3x - 1)^2 + (3y)^2 = a^2 - b^2$ (b) $(3x - 1)^2 + (3y)^2 = a^2 + b^2$
 (c) $(3x + 1)^2 + (3y)^2 = a^2 + b^2$ (d) $(3x + 1)^2 + (3y)^2 = a^2 - b^2$
58. The equation of the straight line passing through $(4, 3)$ and making intercepts on co-ordinate axes whose sum is -1 is :
 (a) $\frac{x}{2} + \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$ (b) $\frac{x}{2} - \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$
 (c) $\frac{x}{2} + \frac{y}{3} = 1$ and $\frac{x}{2} + \frac{y}{1} = 1$ (d) $\frac{x}{2} - \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$

59. Let $A \equiv (3, 2)$ and $B \equiv (5, 1)$. ABP is an equilateral triangle is constructed one the side of AB remote from the origin then the orthocentre of triangle ABP is:
- (a) $\left(4 - \frac{1}{2}\sqrt{3}, \frac{3}{2} - \sqrt{3}\right)$ (b) $\left(4 + \frac{1}{2}\sqrt{3}, \frac{3}{2} + \sqrt{3}\right)$
 (c) $\left(4 - \frac{1}{6}\sqrt{3}, \frac{3}{2} - \frac{1}{3}\sqrt{3}\right)$ (d) $\left(4 + \frac{1}{6}\sqrt{3}, \frac{3}{2} + \frac{1}{3}\sqrt{3}\right)$
60. Area of the triangle formed by the lines through point $(6, 0)$ and at a perpendicular distance of 5 from point $(1, 3)$ and line $y = 16$ in square units is :
- (a) 160 (b) 200 (c) 240 (d) 130
61. The straight lines $3x + y - 4 = 0$, $x + 3y - 4 = 0$ and $x + y = 0$ form a triangle which is :
- (a) equilateral (b) right-angled
 (c) acute-angled and isosceles (d) obtuse-angled and isosceles
62. The orthocentre of the triangle with vertices $(5, 0)$, $(0, 0)$, $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$ is :
- (a) $(2, 3)$ (b) $\left(\frac{5}{2}, \frac{5}{2\sqrt{3}}\right)$ (c) $\left(\frac{5}{6}, \frac{5}{2\sqrt{3}}\right)$ (d) $\left(\frac{5}{2}, \frac{5}{\sqrt{3}}\right)$
63. All chords of a curve $3x^2 - y^2 - 2x + 4y = 0$ which subtends a right angle at the origin passes through a fixed point, which is :
- (a) $(1, 2)$ (b) $(1, -2)$ (c) $(2, 1)$ (d) $(-2, 1)$
64. Let $P(-1, 0)$, $Q(0, 0)$, $R(3, 3\sqrt{3})$ be three points then the equation of the bisector of the angle $\angle PQR$ is :
- (a) $\frac{\sqrt{3}}{2}x + y = 0$ (b) $x + \sqrt{3}y = 0$ (c) $\sqrt{3}x + y = 0$ (d) $x + \frac{\sqrt{3}}{2}y = 0$

Answers

1. (b)	2. (c)	3. (c)	4. (c)	5. (c)	6. (c)	7. (b)	8. (b)	9. (b)	10. (a)
11. (d)	12. (d)	13. (d)	14. (c)	15. (d)	16. (b)	17. (c)	18. (b)	19. (d)	20. (a)
21. (a)	22. (a)	23. (c)	24. (a)	25. (c)	26. (b)	27. (d)	28. (a)	29. (b)	30. (a)
31. (c)	32. (d)	33. (c)	34. (a)	35. (d)	36. (d)	37. (c)	38. (c)	39. (d)	40. (a)
41. (d)	42. (b)	43. (d)	44. (c)	45. (a)	46. (c)	47. (d)	48. (c)	49. (d)	50. (c)
51. (a)	52. (c)	53. (d)	54. (b)	55. (c)	56. (a)	57. (b)	58. (d)	59. (d)	60. (c)
61. (d)	62. (b)	63. (b)	64. (c)						

Exercise-2 : One or More than One Answer is/are Correct

1. A line makes intercepts on co-ordinate axes whose sum is 9 and their product is 20 ; then its equation is/are :
(a) $4x + 5y - 20 = 0$ (b) $5x + 4y - 20 = 0$
(c) $4x - 5y - 20 = 0$ (d) $4x + 5y + 20 = 0$
2. The equation(s) of the medians of the triangle formed by the points (4, 8), (3, 2) and (5, -6) is/are :
(a) $x = 4$ (b) $x = 5y - 3$
(c) $2x + 3y - 12 = 0$ (d) $22x + 3y - 92 = 0$
3. The value(s) of t for which the lines $2x + 3y = 5$, $t^2x + ty - 6 = 0$ and $3x - 2y - 1 = 0$ are concurrent, can be :
(a) $t = 2$ (b) $t = -3$
(c) $t = -2$ (d) $t = 3$
4. If one of the lines given by the equation $ax^2 + 6xy + by^2 = 0$ bisects the angle between the co-ordinate axes, then value of $(a + b)$ can be :
(a) -6 (b) 3 (c) 6 (d) 12
5. Suppose $ABCD$ is a quadrilateral such that the coordinates of A , B and C are (1, 3), (-2, 6) and (5, -8) respectively. For what choices of coordinates of D will make $ABCD$ a trapezium ?
(a) (3, -6) (b) (6, -9) (c) (0, 5) (d) (3, -1)
6. One diagonal of a square is the portion of the line $\sqrt{3}x + y = 2\sqrt{3}$ intercepted by the axes. Then an extremity of the other diagonal is :
(a) $(1 + \sqrt{3}, \sqrt{3} - 1)$ (b) $(1 + \sqrt{3}, \sqrt{3} + 1)$
(c) $(1 - \sqrt{3}, \sqrt{3} - 1)$ (d) $(1 - \sqrt{3}, \sqrt{3} + 1)$
7. Two sides of a rhombus $ABCD$ are parallel to lines $y = x + 2$ and $y = 7x + 3$. If the diagonals of the rhombus intersect at point (1, 2) and the vertex A is on the y -axis is, then the possible coordinates of A are:
(a) $(0, \frac{5}{2})$ (b) (0, 0) (c) (0, 5) (d) (0, 3)
8. The equation of the sides of the triangle having (3, -1) as a vertex and $x - 4y + 10 = 0$ and $6x + 10y - 59 = 0$ as angle bisector and as median respectively drawn from different vertices, are :
(a) $6x + 7y - 13 = 0$ (b) $2x + 9y - 65 = 0$
(c) $18x + 13y - 41 = 0$ (d) $6x - 7y - 25 = 0$
9. $A(1, 3)$ and $C(5, 1)$ are two opposite vertices of a rectangle $ABCD$. If the slope of BD is 2, then the coordinates of B can be :
(a) (4, 4) (b) (5, 4)
(c) (2, 0) (d) (1, 0)

10. All the points lying inside the triangle formed by the points (1, 3), (5, 6), and (-1, 2) satisfy:
- (a) $3x + 2y \geq 0$ (b) $2x + y + 1 \geq 0$
 (c) $-2x + 11 \geq 0$ (d) $2x + 3y - 12 \geq 0$
11. The slope of a median, drawn from the vertex A of the triangle ABC is -2. The co-ordinates of vertices B and C are respectively (-1, 3) and (3, 5). If the area of the triangle be 5 square units, then possible distance of vertex A from the origin is/are.
- (a) 6 (b) 4 (c) $2\sqrt{2}$ (d) $3\sqrt{2}$
12. The points A(0, 0), B(cos α , sin α) and C(cos β , sin β) are the vertices of a right angled triangle if:
- (a) $\sin\left(\frac{\alpha - \beta}{2}\right) = \frac{1}{\sqrt{2}}$ (b) $\cos\left(\frac{\alpha - \beta}{2}\right) = -\frac{1}{\sqrt{2}}$
 (c) $\cos\left(\frac{\alpha - \beta}{2}\right) = \frac{1}{\sqrt{2}}$ (d) $\sin\left(\frac{\alpha - \beta}{2}\right) = -\frac{1}{\sqrt{2}}$

Answers

1.	(a, b)	2.	(a, c, d)	3.	(a, b)	4.	(a, c)	5.	(b, d)	6.	(b, c)
7.	(a, b)	8.	(b, c, d)	9.	(a, c)	10.	(a, b, c, d)	11.	(a, c)	12.	(a, b, c)

Paragraph for Question Nos. 1 to 2

1. Let the equation of BC is $x + py = q$. Then the value of $(p + q)$ if P be the centroid of the ΔABC is:

- (a) 14 (b) -14 (c) 22 (d) -22
2. If P be the orthocentre of $\triangle ABC$ then equation of side BC is :
- (a) $y + 5 = 0$ (b) $y - 5 = 0$ (c) $5y + 1 = 0$ (d) $5y - 1 = 0$

Consider a triangle ABC with vertex $A(2, -4)$. The internal bisectors of the angles B and C are $x + y = 2$ and $x - 3y = 6$ respectively. Let the two bisectors meet at I .

3. If (a, b) is incentre of the triangle ABC then $(a + b)$ has the value equal to :
 (a) 1 (b) 2 (c) 3 (d) 4
4. If (x_1, y_1) and (x_2, y_2) are the co-ordinates of the point B and C respectively, then the value of $(x_1 x_2 + y_1 y_2)$ is equal to :
 (a) 4 (b) 5 (c) 6 (d) 8

Answers

1.	(d)	2.	(a)	3.	(b)	4.	(d)								
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Exercise-4 : Matching Type Problems

1.

Column-I		Column-II	
(A)	If a, b, c are in A.P, then lines $ax + by + c = 0$ are concurrent at:	(P)	$(-4, -7)$
(B)	A point on the line $x + y = 4$ which lies at a unit distance from the line $4x + 3y = 10$ is :	(Q)	$(-7, 11)$
(C)	Orthocentre of triangle made by lines $x + y = 1$, $x - y + 3 = 0$, $2x + y = 7$ is	(R)	$(1, -2)$
(D)	Two vertex of a triangle are $(5, -1)$ and $(-2, 3)$. If orthocentre is the origin then coordinates of the third vertex are	(S)	$(-1, 2)$
		(T)	$(0, 0)$

2.

Column-I		Column-II	
(A)	If $\sum_{r=1}^{n+1} \left(\sum_{k=1}^n {}^k C_{r-1} \right) = 30$, then n is equal to	(P)	1
(B)	The number of integral values of g for which atmost one member of the family of lines given by $(1 + 2\lambda)x + (1 - \lambda)y + 2 + 4\lambda = 0$ (λ is real parameter) is tangent to the circle $x^2 + y^2 + 4gx + 18x + 17y + 4g^2 = 0$ can be	(Q)	4
(C)	Number of solutions of the equation $\sin 9x + \sin 5x + 2\sin^2 x = 1$ in interval $(0, \pi)$ is	(R)	7
(D)	If the roots of the equation $x^2 + ax + b = 0$ ($a, b \in R$) are $\tan 65^\circ$ and $\tan 70^\circ$, then $(a + b)$ equals.	(S)	10

3.

Column-I		Column-II	
(A)	Exact value of $\cos 40^\circ (1 - 2\sin 10^\circ) =$	(P)	$\frac{1}{4}$

(B)	Value of λ for which lines are concurrent $x + y + 1 = 0$, $3x + 2\lambda y + 4 = 0$, $x + y - 3\lambda = 0$ can be	(Q)	$\frac{1}{2}$
(C)	Points $(k, 2 - 2k)$, $(-k + 1, 2k)$ and $(-4 - k, 6 - 2k)$ are collinear then sum of all possible real values of 'k' is	(R)	$\frac{3}{2}$
(D)	Value of $\sum_{k=3}^{\infty} \sin^k\left(\frac{\pi}{6}\right) =$	(S)	$-\frac{1}{2}$

Answers

1. $A \rightarrow R$; $B \rightarrow Q$; $C \rightarrow S$; $D \rightarrow P$
2. $A \rightarrow Q$; $B \rightarrow R$; $C \rightarrow S$; $D \rightarrow P$
3. $A \rightarrow Q$; $B \rightarrow R$; $C \rightarrow S$; $D \rightarrow P$


Exercise-5 : Subjective Type Problems

1. If the area of the quadrilateral $ABCD$ whose vertices are $A(1, 1)$, $B(7, -3)$, $C(12, 2)$ and $D(7, 21)$ is Δ . Find the sum of the digits of Δ .
2. The equation of a line through the mid-point of the sides AB and AD of rhombus $ABCD$, whose one diagonal is $3x - 4y + 5 = 0$ and one vertex is $A(3, 1)$ is $ax + by + c = 0$. Find the absolute value of $(a + b + c)$ where a, b, c are integers expressed in lowest form.
3. If the point (α, α^4) lies on or inside the triangle formed by lines $x^2y + xy^2 - 2xy = 0$, then the largest value of α is.
4. The minimum value of $[(x_1 - x_2)^2 + (12 - \sqrt{1 - x_1^2} - \sqrt{4x_2})^2]^{1/2}$ for all permissible values of x_1 and x_2 is equal to $a\sqrt{b} - c$ where $a, b, c \in \mathbb{N}$, then find the value of $a + b - c$.
5. The number of lines that can be drawn passing through point $(2, 3)$ so that its perpendicular distance from $(-1, 6)$ is equal to 6 is :
6. The graph of $x^4 = x^2y^2$ is a union of n different lines, then the value of n is.
7. The orthocentre of triangle formed by lines $x + y - 1 = 0$, $2x + y - 1 = 0$ and $y = 0$ is (h, k) , then $\frac{1}{k^2} =$
8. Find the integral value of a for which the point $(-2, a)$ lies in the interior of the triangle formed by the lines $y = x$, $y = -x$ and $2x + 3y = 6$.
9. Let $A = (-1, 0)$, $B = (3, 0)$ and PQ be any line passing through $(4, 1)$. The range of the slope of PQ for which there are two points on PQ at which AB subtends a right angle is (λ_1, λ_2) , then $5(\lambda_1 + \lambda_2)$ is equal to.
10. Given that the three points where the curve $y = bx^2 - 2$ intersects the x -axis and y -axis form an equilateral triangle. Find the value of $2b$.

Answers

1.	6	2.	1	3.	1	4.	8	5.	0	6.	3	7.	4
8.	3	9.	6	10.	5								



Exercise-1 : Single Choice Problems

- The locus of mid-points of the chords of the circle $x^2 - 2x + y^2 - 2y + 1 = 0$ which are of unit length is :

(a) $(x-1)^2 + (y-1)^2 = \frac{3}{4}$	(b) $(x-1)^2 + (y-1)^2 = 2$
(c) $(x-1)^2 + (y-1)^2 = \frac{1}{4}$	(d) $(x-1)^2 + (y-1)^2 = \frac{2}{3}$
- The length of a common internal tangent to two circles is 5 and a common external tangent is 15, then the product of the radii of the two circles is :

(a) 25	(b) 50	(c) 75	(d) 30
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- A circle with center (2, 2) touches the coordinate axes and a straight line AB where A and B lie on positive direction of coordinate axes such that the circle lies between origin and the line AB . If O be the origin then the locus of circumcenter of $\triangle OAB$ will be:

(a) $xy = x + y + \sqrt{x^2 + y^2}$	(b) $xy = x + y - \sqrt{x^2 + y^2}$
(c) $xy + x + y = \sqrt{x^2 + y^2}$	(d) $xy + x + y + \sqrt{x^2 + y^2} = 0$
- Length of chord of contact of point (4, 4) with respect to the circle $x^2 + y^2 - 2x - 2y - 7 = 0$ is:

(a) $\frac{3}{\sqrt{2}}$	(b) $3\sqrt{2}$	(c) 3	(d) 6
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- Let P, Q, R, S be the feet of the perpendiculars drawn from a point (1, 1) upon the lines $x + 4y = 12$; $x - 4y + 4 = 0$ and their angle bisectors respectively; then equation of the circle which passes through Q, R, S is :

(a) $x^2 + y^2 - 5x + 3y - 6 = 0$	(b) $x^2 + y^2 - 5x - 3y + 6 = 0$
(c) $x^2 + y^2 - 5x - 3y - 6 = 0$	(d) None of these

6. From a point 'P' on the line $2x + y + 4 = 0$; which is nearest to the circle $x^2 + y^2 - 12y + 35 = 0$, tangents are drawn to given circle. The area of quadrilateral PACB (where 'C' is the center of circle and PA & PB are the tangents.) is :
 (a) 8 (b) $\sqrt{110}$ (c) $\sqrt{19}$ (d) None of these
7. The line $2x - y + 1 = 0$ is tangent to the circle at the point (2, 5) and the centre of the circle lies on $x - 2y = 4$. The radius of the circle is:
 (a) $3\sqrt{5}$ (b) $5\sqrt{3}$
 (c) $2\sqrt{5}$ (d) $5\sqrt{2}$
8. If $A(\cos \alpha, \sin \alpha)$, $B(\sin \alpha, -\cos \alpha)$, $C(1, 2)$ are the vertices of a triangle, then as α varies the locus of centroid of the $\triangle ABC$ is a circle whose radius is :
 (a) $\frac{2\sqrt{2}}{3}$ (b) $\frac{\sqrt{4}}{3}$ (c) $\frac{2}{3}$ (d) $\frac{\sqrt{2}}{9}$
9. Tangents drawn to circle $(x - 1)^2 + (y - 1)^2 = 5$ at point P meets the line $2x + y + 6 = 0$ at Q on the x-axis. Length PQ is equal to :
 (a) $\sqrt{12}$ (b) $\sqrt{10}$ (c) 4 (d) $\sqrt{15}$
10. ABCD is square in which A lies on positive y-axis and B lies on the positive x-axis. If D is the point (12, 17), then co-ordinate of C is :
 (a) (17, 12) (b) (17, 5) (c) (17, 16) (d) (15, 3)
11. **Statement-1:** The lines $y = mx + 1 - m$ for all values of m is a normal to the circle $x^2 + y^2 - 2x - 2y = 0$.
Statement-2: The line L passes through the centre of the circle.
 (a) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (b) Statement-1 is true, statement-2 is true and statement-2 is not the correct explanation for statement-1.
 (c) Statement-1 is true, statement-2 is false.
 (d) Statement-1 is false, statement-2 is true.
12. A(1, 0) and B(0, 1) are two fixed points on the circle $x^2 + y^2 = 1$. C is a variable point on this circle. As C moves, the locus of the orthocentre of the triangle ABC is :
 (a) $x^2 + y^2 - 2x - 2y + 1 = 0$ (b) $x^2 + y^2 - x - y = 0$
 (c) $x^2 + y^2 = 4$ (d) $x^2 + y^2 + 2x - 2y + 1 = 0$
13. Equation of a circle passing through (1, 2) and (2, 1) and for which line $x + y = 2$ is a diameter ; is :
 (a) $x^2 + y^2 + 2x + 2y - 11 = 0$ (b) $x^2 + y^2 - 2x - 2y - 1 = 0$
 (c) $x^2 + y^2 - 2x - 2y + 1 = 0$ (d) None of these

14. The area of an equilateral triangle inscribed in a circle of radius 4 cm, is :
(a) 12 cm^2 (b) $9\sqrt{3} \text{ cm}^2$
(c) $8\sqrt{3} \text{ cm}^2$ (d) $12\sqrt{3} \text{ cm}^2$
15. Let all the points on the curve $x^2 + y^2 - 10x = 0$ are reflected about the line $y = x + 3$. The locus of the reflected points is in the form $x^2 + y^2 + gx + fy + c = 0$. The value of $(g + f + c)$ is equal to :
(a) 28 (b) -28 (c) 38 (d) -38
16. The shortest distance from the line $3x + 4y = 25$ to the circle $x^2 + y^2 = 6x - 8y$ is equal to:
(a) $7/5$ (b) $9/5$ (c) $11/5$ (d) $32/5$
17. In the xy -plane, the length of the shortest path from $(0, 0)$ to $(12, 16)$ that does not go inside the circle $(x - 6)^2 + (y - 8)^2 = 25$ is:
(a) $10\sqrt{3}$ (b) $10\sqrt{5}$
(c) $10\sqrt{3} + \frac{5\pi}{3}$ (d) $10 + 5\pi$
18. A circle is inscribed in an equilateral triangle with side lengths 6 unit. Another circle is drawn inside the triangle (but outside the first circle), tangent to the first circle and two of the sides of the triangle. The radius of the smaller circle is:
(a) $1/\sqrt{3}$ (b) $2/3$
(c) $1/2$ (d) 1
19. The equation of the tangent to the circle $x^2 + y^2 - 4x = 0$ which is perpendicular to the normal drawn through the origin can be :
(a) $x = 1$ (b) $x = 2$ (c) $x + y = 2$ (d) $x = 4$
20. The equation of the line parallel to the line $3x + 4y = 0$ and touching the circle $x^2 + y^2 = 9$ in the first quadrant is :
(a) $3x + 4y = 15$ (b) $3x + 4y = 45$
(c) $3x + 4y = 9$ (d) $3x + 4y = 12$
21. The centres of the three circles $x^2 + y^2 - 10x + 9 = 0$, $x^2 + y^2 - 6x + 2y + 1 = 0$, $x^2 + y^2 - 9x - 4y + 2 = 0$
(a) lie on the straight line $x - 2y = 5$ (b) lie on circle $x^2 + y^2 = 25$
(c) do not lie on straight line (d) lie on circle $x^2 + y^2 + x + y - 17 = 0$
22. The equation of the diameter of the circle $x^2 + y^2 + 2x - 4y = 4$ that is parallel to $3x + 5y = 4$ is:
(a) $3x + 5y = -7$ (b) $3x + 5y = 7$
(c) $3x + 5y = 9$ (d) $3x + 5y = 1$

23. There are two circles passing through points $A(-1, 2)$ and $B(2, 3)$ having radius $\sqrt{5}$. Then the length of intercept on x -axis of the circle intersecting x -axis is :
 (a) 2 (b) 3 (c) 4 (d) 5
24. A square $OABC$ is formed by line pairs $xy = 0$ and $xy + 1 = x + y$ where ' O ' is the origin. A circle with centre C_1 inside the square is drawn to touch the line pair $xy = 0$ and another circle with centre C_2 and radius twice that of C_1 , is drawn to touch the circle C_1 and the other line pair. The radius of the circle with centre C_1 is:
 (a) $\frac{\sqrt{2}}{\sqrt{3}(\sqrt{2}+1)}$ (b) $\frac{2\sqrt{2}}{3(\sqrt{2}+1)}$
 (c) $\frac{\sqrt{2}}{3(\sqrt{2}+1)}$ (d) $\frac{\sqrt{2}+1}{3\sqrt{2}}$
25. The equation of the circle circumscribing the triangle formed by the points $(3, 4)$, $(1, 4)$ and $(3, 2)$ is :
 (a) $8x^2 + 8y^2 - 16x - 13y = 0$ (b) $x^2 + y^2 - 4x - 8y + 19 = 0$
 (c) $x^2 + y^2 - 4x - 6y + 11 = 0$ (d) $x^2 + y^2 - 6x - 6y + 17 = 0$
26. The equation of the tangent to circle $x^2 + y^2 + 2gx + 2fy = 0$ at the origin is :
 (a) $fx + gy = 0$ (b) $gx + fy = 0$ (c) $x = 0$ (d) $y = 0$
27. The line $y = x$ is tangent at $(0, 0)$ to a circle of radius 1. The centre of the circle is :
 (a) either $\left(-\frac{1}{2}, \frac{1}{2}\right)$ or $\left(\frac{1}{2}, -\frac{1}{2}\right)$ (b) either $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ or $\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$
 (c) either $\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ or $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (d) either $(1, 0)$ or $(-1, 0)$
28. The circles $x^2 + y^2 + 6x + 6y = 0$ and $x^2 + y^2 - 12x - 12y = 0$:
 (a) cut orthogonally (b) touch each other internally
 (c) intersect in two points (d) touch each other externally
29. In a right triangle ABC , right angled at A , on the leg AC as diameter, a semicircle is described. The chord joining A with the point of intersection D of the hypotenuse and the semicircle, then the length AC equals to:
 (a) $\frac{AB \cdot AD}{\sqrt{AB^2 + AD^2}}$ (b) $\frac{AB \cdot AD}{AB + AD}$
 (c) $\sqrt{AB \cdot AD}$ (d) $\frac{AB \cdot AD}{\sqrt{AB^2 - AD^2}}$
30. Radical centre of the circles drawn on the sides as a diameter of triangle formed by the lines $3x - 4y + 6 = 0$, $x - y + 2 = 0$ and $4x + 3y - 17 = 0$ is :
 (a) $(3, 2)$ (b) $(3, -2)$ (c) $(2, -3)$ (d) $(2, 3)$

- 31. Statement-1:** A circle can be inscribed in a quadrilateral whose sides are $3x - 4y = 0$, $3x - 4y = 5$, $3x + 4y = 0$ and $3x + 4y = 7$.
- Statement-2:** A circle can be inscribed in a parallelogram if and only if it is a rhombus.
- (a) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (b) Statement-1 is true, statement-2 is true and statement-2 is not the correct explanation for statement-1.
 (c) Statement-1 is true, statement-2 is false.
 (d) Statement-1 is false, statement-2 is true.
- 32.** If $x = 3$ is the chord of contact of the circle $x^2 + y^2 = 81$, then the equation of the corresponding pair of tangents, is:
 (a) $x^2 - 8y^2 + 54x + 729 = 0$ (b) $x^2 - 8y^2 - 54x + 729 = 0$
 (c) $x^2 - 8y^2 - 54x - 729 = 0$ (d) $x^2 - 8y^2 = 729$
- 33.** The shortest distance from the line $3x + 4y = 25$ to the circle $x^2 + y^2 = 6x - 8y$ is equal to :
 (a) $\frac{7}{3}$ (b) $\frac{9}{5}$ (c) $\frac{11}{5}$ (d) $\frac{7}{5}$
- 34.** The circle with equation $x^2 + y^2 = 1$ intersects the line $y = 7x + 5$ at two distinct points A and B. Let C be the point at which the positive x-axis intersects the circle. The angle ACB is :
 (a) $\tan^{-1} \frac{4}{3}$ (b) $\cot^{-1}(-1)$ (c) $\tan^{-1} 1$ (d) $\cot^{-1} \frac{4}{3}$
- 35.** The abscissae of two points A and B are the roots of the equation $x^2 + 2ax - b^2 = 0$ and their ordinates are the roots of the equation $x^2 + 2px - q^2 = 0$. The radius of the circle with AB as diameter is ::
 (a) $\sqrt{a^2 + b^2 + p^2 + q^2}$ (b) $\sqrt{a^2 + p^2}$
 (c) $\sqrt{b^2 + q^2}$ (d) $\sqrt{a^2 + b^2 + p^2 + 1}$
- 36.** Let C be the circle of radius unity centred at the origin. If two positive numbers x_1 and x_2 are such that the line passing through $(x_1, -1)$ and $(x_2, 1)$ is tangent to C then:
 (a) $x_1 x_2 = 1$ (b) $x_1 x_2 = -1$
 (c) $x_1 + x_2 = 1$ (d) $4x_1 x_2 = 1$
- 37.** A circle bisects the circumference of the circle $x^2 + y^2 + 2y - 3 = 0$ and touches the line $x = y$ at the point $(1, 1)$. Its radius is :
 (a) $\frac{3}{\sqrt{2}}$ (b) $\frac{9}{\sqrt{2}}$ (c) $4\sqrt{2}$ (d) $3\sqrt{2}$
- 38.** The distance between the chords of contact of tangents to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ from the origin and the point (g, f) is:

- (a) $\sqrt{g^2 + f^2}$ (b) $\frac{\sqrt{g^2 + f^2 - c}}{2}$
 (c) $\frac{g^2 + f^2 - c}{2\sqrt{g^2 + f^2}}$ (d) $\frac{\sqrt{g^2 + f^2 + c}}{2\sqrt{g^2 + f^2}}$

39. If the tangents AP and AQ are drawn from the point $A(3, -1)$ to the circle $x^2 + y^2 - 3x + 2y - 7 = 0$ and C is the centre of circle, then the area of quadrilateral $APCQ$ is :
 (a) 9 (b) 4 (c) 2 (d) non-existent
40. Number of integral value(s) of k for which no tangent can be drawn from the point $(k, k + 2)$ to the circle $x^2 + y^2 = 4$ is :
 (a) 0 (b) 1 (c) 2 (d) 3
41. If the length of the normal for each point on a curve is equal to the radius vector, then the curve :
 (a) is a circle passing through origin
 (b) is a circle having centre at origin and radius > 0
 (c) is a circle having centre on x -axis and touching y -axis
 (d) is a circle having centre on y -axis and touching x -axis
42. A circle of radius unity is centred at origin. Two particles start moving at the same time from the point $(1, 0)$ and move around the circle in opposite direction. One of the particle moves counter clockwise with constant speed v and the other moves clockwise with constant speed $3v$. After leaving $(1, 0)$, the two particles meet first at a point P and continue until they meet next at point Q . The coordinates of the point Q are:
 (a) $(1, 0)$ (b) $(0, 1)$
 (c) $(0, -1)$ (d) $(-1, 0)$
43. A variable circle is drawn to touch the x -axis at the origin. The locus of the pole of the straight line $lx + my + n = 0$ w.r.t the variable circle has the equation:
 (a) $x(my - n) - ly^2 = 0$ (b) $x(my + n) - ly^2 = 0$
 (c) $x(my - n) + ly^2 = 0$ (d) none of these
44. The minimum length of the chord of the circle $x^2 + y^2 + 2x + 2y - 7 = 0$ which is passing through $(1, 0)$ is :
 (a) 2 (b) 4 (c) $2\sqrt{2}$ (d) $\sqrt{5}$
45. Three concentric circles of which the biggest is $x^2 + y^2 = 1$, have their radii in A.P. If the line $y = x + 1$ cuts all the circles in real and distinct points. The interval in which the common difference of the A.P. will lie is:
 (a) $\left(0, \frac{1}{4}\right)$ (b) $\left(0, \frac{1}{2\sqrt{2}}\right)$ (c) $\left(0, \frac{2 - \sqrt{2}}{4}\right)$ (d) none

46. The locus of the point of intersection of the tangent to the circle $x^2 + y^2 = a^2$, which include an angle of 45° is the curve $(x^2 + y^2)^2 = \lambda a^2(x^2 + y^2 - a^2)$. The value of λ is:
 (a) 2 (b) 4
 (c) 8 (d) 16
47. A circle touches the line $y = x$ at point $(4, 4)$ on it. The length of the chord on the line $x + y = 0$ is $6\sqrt{2}$. Then one of the possible equation of the circle is :
 (a) $x^2 + y^2 + x - y + 30 = 0$ (b) $x^2 + y^2 + 2x - 18y + 32 = 0$
 (c) $x^2 + y^2 + 2x + 18y + 32 = 0$ (d) $x^2 + y^2 - 2x - 22y + 32 = 0$
48. Point on the circle $x^2 + y^2 - 2x + 4y - 4 = 0$ which is nearest to the line $y = 2x + 11$ is :
 (a) $\left(1 - \frac{6}{\sqrt{5}}, -2 + \frac{3}{\sqrt{5}}\right)$ (b) $\left(1 + \frac{6}{\sqrt{5}}, -2 - \frac{3}{\sqrt{5}}\right)$
 (c) $\left(1 - \frac{6}{\sqrt{5}}, -2 - \frac{3}{\sqrt{5}}\right)$ (d) None of these
49. A foot of the normal from the point $(4, 3)$ to a circle is $(2, 1)$ and a diameter of the circle has the equation $2x - y - 2 = 0$. Then the equation of the circle is:
 (a) $x^2 + y^2 - 4y + 2 = 0$ (b) $x^2 + y^2 - 4y + 1 = 0$
 (c) $x^2 + y^2 - 2x - 1 = 0$ (d) $x^2 + y^2 - 2x + 1 = 0$
50. If $\left(a, \frac{1}{a}\right), \left(b, \frac{1}{b}\right), \left(c, \frac{1}{c}\right)$ and $\left(d, \frac{1}{d}\right)$ are four distinct points on a circle of radius 4 units then, $abcd$ is equal to:
 (a) 4 (b) $1/4$ (c) 1 (d) 16

Answers

1. (a)	2. (b)	3. (a)	4. (b)	5. (b)	6. (c)	7. (a)	8. (d)	9. (a)	10. (b)
11. (a)	12. (a)	13. (c)	14. (d)	15. (c)	16. (a)	17. (c)	18. (a)	19. (d)	20. (a)
21. (c)	22. (b)	23. (c)	24. (c)	25. (c)	26. (b)	27. (c)	28. (d)	29. (d)	30. (d)
31. (d)	32. (b)	33. (d)	34. (c)	35. (a)	36. (a)	37. (b)	38. (c)	39. (d)	40. (b)
41. (b)	42. (d)	43. (a)	44. (b)	45. (c)	46. (c)	47. (b)	48. (a)	49. (c)	50. (c)


Exercise-2 : One or More than One Answer is/are Correct

- Number of circle touching both the axes and the line $x + y = 4$ is greater than or equal to :
 (a) 1 (b) 2
 (c) 3 (d) 4
- Which of the following is/are true ?
 The circles $x^2 + y^2 - 6x - 6y + 9 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$ are such that :
 (a) They do not intersect
 (b) They touch each other
 (c) Their exterior common tangents are parallel
 (d) Their interior common tangents are perpendicular
- Let ' α ' be a variable parameter, then the length of the chord of the curve :
 $(x - \sin^{-1} \alpha)(x - \cos^{-1} \alpha) + (y - \sin^{-1} \alpha)(y + \cos^{-1} \alpha) = 0$
 along the line $x = \frac{\pi}{4}$ can not be equal to :
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$
- If the point $(1, 4)$ lies inside the circle $x^2 + y^2 - 6x - 10y + p = 0$ and the circle does not touch or intersect the coordinate axes, then which of the following must be correct :
 (a) $p < 29$ (b) $p > 25$
 (c) $p > 27$ (d) $p < 27$
- The equation of a circle $S_1 = 0$ is $x^2 + y^2 = 4$, locus of the intersection of orthogonal tangents to the circle is the curve C_1 and the locus of the intersection of perpendicular tangents to the curve C_1 is the curve C_2 , then :
 (a) C_2 is a circle
 (b) C_1, C_2 are circles having different centres
 (c) C_1, C_2 are circles having same centres
 (d) area enclosed between C_1 and C_2 is 8π
- If two distinct chords drawn from the point (p, q) on the circle $x^2 + y^2 = px + qy$ (where $pq \neq 0$) are bisected by the x -axis, then :
 (a) $p^2 = q^2$ (b) $p^2 > q^2$
 (c) $p^2 < 8q^2$ (d) $p^2 > 8q^2$
- If $a = \max\{(x+2)^2 + (y-3)^2\}$ and $b = \min\{(x+2)^2 + (y-3)^2\}$ where x, y satisfying $x^2 + y^2 + 8x - 10y - 40 = 0$, then :
 (a) $a + b = 18$ (b) $a + b = 178$ (c) $a - b = 4\sqrt{2}$ (d) $a - b = 72\sqrt{2}$

8. The locus of points of intersection of the tangents to $x^2 + y^2 = a^2$ at the extremities of a chord of circle $x^2 + y^2 = a^2$ which touches the circle $x^2 + y^2 - 2ax = 0$ is/are :
- (a) $y^2 = a(a - 2x)$ (b) $x^2 = a(a - 2y)$
 (c) $x^2 + y^2 = (x - a)^2$ (d) $x^2 + y^2 = (y - a)^2$
9. A circle passes through the points $(-1, 1)$, $(0, 6)$ and $(5, 5)$. The point(s) on this circle, the tangent(s) at which is/are parallel to the straight line joining the origin to its centre is/are
 (a) $(1, -5)$ (b) $(5, 1)$ (c) $(-5, -1)$ (d) $(-1, 5)$
10. A square is inscribed in the circle $x^2 + y^2 - 2x + 4y - 93 = 0$ with the sides parallel to the co-ordinate axes. The co-ordinate of the vertices are :
 (a) $(8, 5)$ (b) $(8, 9)$ (c) $(-6, 5)$ (d) $(-6, -9)$

Answers

1.	(a, b, c, d)	2.	(a, c, d)	3.	(a, b, c)	4.	(a, b)	5.	(a, c, d)	6.	(b, d)
7.	(b, d)	8.	(a, c)	9.	(b, d)	10.	(a, c)				

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 3

Let each of the circles,

$$S_1 \equiv x^2 + y^2 + 4y - 1 = 0,$$

$$S_2 \equiv x^2 + y^2 + 6x + y + 8 = 0,$$

$$S_3 \equiv x^2 + y^2 - 4x - 4y - 37 = 0$$

touches the other two. Let P_1, P_2, P_3 be the points of contact of S_1 and S_2 , S_2 and S_3 , S_3 and S_1 respectively and C_1, C_2, C_3 be the centres of S_1, S_2, S_3 respectively.

- The co-ordinates of P_1 are :
 (a) $(2, -1)$ (b) $(2, 1)$ (c) $(-2, 1)$ (d) $(-2, -1)$
- The ratio $\frac{\text{area}(\Delta P_1 P_2 P_3)}{\text{area}(\Delta C_1 C_2 C_3)}$ is equal to :
 (a) $3:2$ (b) $2:5$ (c) $5:3$ (d) $2:3$
- P_2 and P_3 are image of each other with respect to line :
 (a) $y = x + 1$ (b) $y = -x$ (c) $y = x$ (d) $y = -x + 2$

Paragraph for Question Nos. 4 to 6

Let $A(3, 7)$ and $B(6, 5)$ are two points. $C : x^2 + y^2 - 4x - 6y - 3 = 0$ is a circle.

- The chords in which the circle C cuts the members of the family S of circle passing through A and B are concurrent at :
 (a) $(2, 3)$ (b) $\left(2, \frac{23}{3}\right)$ (c) $\left(3, \frac{23}{2}\right)$ (d) $(3, 2)$
- Equation of the member of the family of circles S that bisects the circumference of C is :
 (a) $x^2 + y^2 - 5x - 1 = 0$ (b) $x^2 + y^2 - 5x + 6y - 1 = 0$
 (c) $x^2 + y^2 - 5x - 6y - 1 = 0$ (d) $x^2 + y^2 + 5x - 6y - 1 = 0$
- If O is the origin and P is the center of C , then absolute value of difference of the squares of the lengths of the tangents from A and B to the circle C is equal to :
 (a) $(AB)^2$ (b) $(OP)^2$ (c) $|(AP)^2 - (BP)^2|$ (d) $(AP)^2 + (BP)^2$

Paragraph for Question Nos. 7 to 8

Let the diameter of a subset S of the plane be defined as the maximum of the distance between arbitrary pairs of points of S .

- Let $S = \{(x, y) : (y - x) \leq 0, x + y \geq 0, x^2 + y^2 \leq 2\}$, then the diameter of S is :
 (a) 2 (b) 4 (c) $\sqrt{2}$ (d) $2\sqrt{2}$

8. Let $S = \{(x, y) : (\sqrt{5} - 1)x - \sqrt{10 + 2\sqrt{5}} y \geq 0, (\sqrt{5} - 1)x + \sqrt{10 + 12\sqrt{5}} y \geq 0, x^2 + y^2 \leq 9\}$ then the diameter of S is :

- (a) $\frac{3}{2}(\sqrt{5} - 1)$ (b) $3(\sqrt{5} - 1)$ (c) $3\sqrt{2}$ (d) 3

Paragraph for Question Nos. 9 to 10

Let L_1, L_2 and L_3 be the lengths of tangents drawn from a point P to the circles $x^2 + y^2 = 4$, $x^2 + y^2 - 4x = 0$ and $x^2 + y^2 - 4y = 0$ respectively. If $L_1^4 = L_2^2 L_3^2 + 16$ then the locus of P are the curves, C_1 (a straight line) and C_2 (a circle).

9. Circum centre of the triangle formed by C_1 and two other lines which are at angle of 45° with C_1 and tangent to C_2 is :
 (a) (1, 1) (b) (0, 0) (c) (-1, -1) (d) (2, 2)
10. If S_1, S_2 and S_3 are three circles congruent to C_2 and touch both C_1 and C_2 ; then the area of triangle formed by joining centres of the circles S_1, S_2 and S_3 is (in square units)
 (a) 2 (b) 4 (c) 8 (d) 16

Answers

1.	(d)	2.	(b)	3.	(c)	4.	(b)	5.	(c)	6.	(c)	7.	(a)	8.	(d)	9.	(b)	10.	(c)
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Exercise-4 : Matching Type Problems

1.

Column-I		Column-II	
(A)	The triangle PQR is inscribed in the circle $x^2 + y^2 = 169$. If $Q(5, 12)$ and $R(-12, 5)$ then $\angle QPR$ is	(P)	$\pi/6$
(B)	The angle between the lines joining the origin to the points of intersection of the line $4x + 3y = 24$ with circle $(x - 3)^2 + (y - 4)^2 = 25$	(Q)	$\pi/4$
(C)	Two parallel tangents drawn to given circle are cut by a third tangent. The angle subtended by the portion of third tangent between the given tangents at the centre is	(R)	$\pi/3$
(D)	A chord is drawn joining the point of contact of tangents drawn from a point P to the circle. If the chord subtends an angle $\pi/2$ at the centre then the angle included between the tangents at P is	(S)	$\pi/2$
		(T)	π

2.

Column-I		Column-II	
(A)	A ray of light coming from the point $(1, 2)$ is reflected at a point A on the x -axis then passes through the point $(5, 3)$. The coordinates of the point A are :	(P)	$\left(\frac{13}{5}, 0\right)$
(B)	The equation of three sides of triangle ABC are $x + y = 3$, $x - y = 5$ and $3x + y = 4$. Considering the sides as diameter, three circles S_1, S_2, S_3 are drawn whose radical centre is at :	(Q)	$(4, -1)$
(C)	If the straight line $x - 2y + 1 = 0$ intersects the circle $x^2 + y^2 = 25$ at the points P and Q , then the coordinate of the point of intersection of tangents drawn at P and Q to the circle is	(R)	$(-25, 50)$
(D)	The equation of three sides of a triangle are $4x + 3y + 9 = 0$, $2x + 3 = 0$ and $3y - 4 = 0$. The circumcentre of the triangle is :	(S)	$\left(\frac{-19}{8}, \frac{1}{6}\right)$
		(T)	$(-1, 2)$

Answers

1. $A \rightarrow Q; B \rightarrow S; C \rightarrow S; D \rightarrow S$ 2. $A \rightarrow P; B \rightarrow Q; C \rightarrow R; D \rightarrow S$



Exercise-5 : Subjective Type Problems

1. Tangents are drawn to circle $x^2 + y^2 = 1$ at its intersection points (distinct) with the circle $x^2 + y^2 + (\lambda - 3)x + (2\lambda + 2)y + 2 = 0$. The locus of intersection of tangents is a straight line, then the slope of that straight line is.
2. The radical centre of the three circles is at the origin. The equations of the two of the circles are $x^2 + y^2 = 1$ and $x^2 + y^2 + 4x + 4y - 1 = 0$. If the third circle passes through the points $(1, 1)$ and $(-2, 1)$; and its radius can be expressed in the form of $\frac{p}{q}$, where p and q are relatively prime positive integers. Find the value of $(p + q)$.
3. Let $S = \{(x, y) \mid x, y \in \mathbb{R}, x^2 + y^2 - 10x + 16 = 0\}$. The largest value of $\frac{y}{x}$ can be put in the form $\frac{m}{n}$ where m, n are relatively prime natural numbers, then $m^2 + n^2 =$
4. In the above problem, the complete range of the expression $x^2 + y^2 - 26x + 12y + 210$ is $[a, b]$, then $b - 2a =$
5. If the line $y = 2 - x$ is tangent to the circle S at the point $P(1, 1)$ and circle S is orthogonal to the circle $x^2 + y^2 + 2x + 2y - 2 = 0$, then find the length of tangent drawn from the point $(2, 2)$ to circle S .
6. Two circles having radii r_1 and r_2 passing through vertex A of a triangle ABC . One of the circle touches the side BC at B and other circle touches the side BC at C . If $a = 5$ and $A = 30^\circ$, find $\sqrt{r_1 r_2}$.
7. A circle S of radius ' a ' is the director circle of another circle S_1 . S_1 is the director circle of S_2 and so on. If the sum of radius of $S, S_1, S_2, S_3, \dots$ circles is ' 2 ' and $a = (k - \sqrt{k})$, then the value of k is
8. If r_1 and r_2 be the maximum and minimum radius of the circle which pass through the point $(4, 3)$ and touch the circle $x^2 + y^2 = 49$, then $\frac{r_1}{r_2}$ is
9. Let C be the circle $x^2 + y^2 - 4x - 4y - 1 = 0$. The number of points common to C and the sides of the rectangle determined by the lines $x = 2, x = 5, y = -1$ and $y = 5$ is P then find P .
10. Two congruent circles with centres at $(2, 3)$ and $(5, 6)$ intersects at right angle; find the radius of the circle.
11. The sum of abscissa and ordinate of a point on the circle $x^2 + y^2 - 4x + 2y - 20 = 0$ which is nearest to $\left(2, \frac{3}{2}\right)$ is :
12. AB is any chord of the circle $x^2 + y^2 - 6x - 8y - 11 = 0$ which subtends an angle $\frac{\pi}{2}$ at $(1, 2)$. If locus of midpoint of AB is a circle $x^2 + y^2 - 2ax - 2by - c = 0$; then find the value of $(a + b + c)$.

13. If circles $x^2 + y^2 = c$ with radius $\sqrt{3}$ and $x^2 + y^2 + ax + by + c = 0$ with radius $\sqrt{6}$ intersect at two points A and B . If length of $AB = \sqrt{l}$. Find l .

Answers

1.	2	2.	5	3.	25	4.	66	5.	2	6.	5	7.	2
8.	6	9.	3	10.	3	11.	6	12.	8	13.	8		




Exercise-1 : Single Choice Problems

- Let PQ be the latus rectum of the parabola $y^2 = 4x$ with vertex A . Minimum length of the projection of PQ on a tangent drawn in portion of parabola PAQ is :
 (a) 2 (b) 4
 (c) $2\sqrt{3}$ (d) $2\sqrt{2}$
- A normal is drawn to the parabola $y^2 = 9x$ at the point $P(4, 6)$. A circle is described on SP as diameter; where S is the focus. The length of the intercept made by the circle on the normal at point P is :
 (a) $\frac{17}{4}$ (b) $\frac{15}{4}$ (c) 4 (d) 5
- A trapezium is inscribed in the parabola $y^2 = 4x$, such that its diagonal pass through the point $(1, 0)$ and each has length $\frac{25}{4}$. If the area of the trapezium be P , then $4P$ is equal to :
 (a) 70 (b) 71 (c) 80 (d) 75
- The length of normal chord of parabola $y^2 = 4x$, which subtends an angle of 90° at the vertex is :
 (a) $6\sqrt{3}$ (b) $7\sqrt{2}$ (c) $8\sqrt{2}$ (d) $9\sqrt{2}$
- If b and c are the lengths of the segments of any focal chord of a parabola $y^2 = 4ax$. Then the length of semi-latus rectum is :
 (a) $\frac{bc}{b+c}$ (b) $\frac{2bc}{b+c}$
 (c) $\frac{b+c}{2}$ (d) $\sqrt{bc}$
- The length of the shortest path that begins at the point $(-1, 1)$, touches the x -axis and then ends at a point on the parabola $(x-y)^2 = 2(x+y-4)$, is :
 (a) $3\sqrt{2}$ (b) 5 (c) $4\sqrt{10}$ (d) 13

7. If the normals at three points P, Q, R of the parabola $y^2 = 4ax$ meet in a point O' and S be its focus, then $|SP| \cdot |SQ| \cdot |SR|$ is equal to :
 (a) a^3 (b) $a^2(SO')$
 (c) $a(SO')^2$ (d) None of these
8. Let P and Q are points on the parabola $y^2 = 4ax$ with vertex O , such that OP is perpendicular to OQ and have lengths r_1 and r_2 respectively, then the value of $\frac{r_1^{4/3} r_2^{4/3}}{r_1^{2/3} + r_2^{2/3}}$ is :
 (a) $16a^2$ (b) a^2 (c) $4a$ (d) None of these
9. Length of the shortest chord of the parabola $y^2 = 4x + 8$, which belongs to the family of lines $(1 + \lambda)y + (\lambda - 1)x + 2(1 - \lambda) = 0$, is :
 (a) 6 (b) 5 (c) 8 (d) 2
10. If locus of mid-point of any normal chord of the parabola :
 $y^2 = 4x$ is $x - a = \frac{b}{y^2} + \frac{y^2}{c}$;
 where $a, b, c \in N$, then $(a + b + c)$ equals to :
 (a) 5 (b) 8 (c) 10 (d) None of these
11. Let tangents at P and Q to curve $y^2 - 4x - 2y + 5 = 0$ intersect at T . If $S(2, 1)$ is a point such that $(SP)(SQ) = 16$, then the length ST is equal to :
 (a) 3 (b) 4 (c) 5 (d) None of these
12. Abscissa of two points P and Q on parabola $y^2 = 8x$ are roots of equation $x^2 - 17x + 11 = 0$. Let Tangents at P and Q meet at point T , then distance of T from the focus of parabola is :
 (a) 7 (b) 6 (c) 5 (d) 4
13. If $Ax + By = 1$ is a normal to the curve $ay = x^2$, then :
 (a) $4A^2(1 - aB) = aB^3$ (b) $4A^2(2 + aB) = aB^3$
 (c) $4A^2(1 + aB) + aB^3 = 0$ (d) $2A^2(2 - aB) = aB^3$
14. The equation of a curve which passes through the point $(3, 1)$, such that the segment of any tangent between the point of tangency and the x -axis is bisected at its point of intersection with y -axis, is :
 (a) $x = 3y^2$ (b) $x^2 = 9y$ (c) $x = y^2 + 2$ (d) $2x = 3y^2 + 3$
15. The parabola $y = 4 - x^2$ has vertex P . It intersects x -axis at A and B . If the parabola is translated from its initial position to a new position by moving its vertex along the line $y = x + 4$, so that it intersects x -axis at B and C , then abscissa of C will be :
 (a) 3 (b) 4 (c) 6 (d) 8

16. A focal chord for parabola $y^2 = 8(x + 2)$ is inclined at an angle of 60° with positive x -axis and intersects the parabola at P and Q . Let perpendicular bisector of the chord PQ intersects the x -axis at R ; then the distance of R from focus is :
- (a) $\frac{8}{3}$ (b) $\frac{16\sqrt{3}}{3}$ (c) $\frac{16}{3}$ (d) $8\sqrt{3}$
17. The Director circle of the parabola $(y - 2)^2 = 16(x + 7)$ touches the circle $(x - 1)^2 + (y + 1)^2 = r^2$, then r is equal to :
- (a) 10 (b) 11 (c) 12 (d) None of these
18. The chord of contact of a point $A(x_A, y_A)$ of $y^2 = 4x$ passes through $(3, 1)$ and point A lies on $x^2 + y^2 = 5^2$. Then :
- (a) $5x_A^2 + 24x_A + 11 = 0$ (b) $13x_A^2 + 8x_A - 21 = 0$
 (c) $5x_A^2 + 24x_A + 61 = 0$ (d) $13x_A^2 + 21x_A - 31 = 0$

Answers

1. (d)	2. (b)	3. (d)	4. (a)	5. (b)	6. (a)	7. (c)	8. (a)	9. (c)	10. (b)
11. (b)	12. (a)	13. (d)	14. (a)	15. (d)	16. (c)	17. (c)	18. (a)		

Paragraph for Question Nos. 1 to 3

Consider the following lines :

$$L_1 : x - y - 1 = 0$$

$$L_2 : x + y - 5 = 0$$

$$L_3 : y - 4 = 0$$

Let L_1 is axis to a parabola, L_2 is tangent at the vertex to this parabola and L_3 is another tangent to this parabola at some point P .

Let 'C' be the circle circumscribing the triangle formed by tangent and normal at point P and axis of parabola. The tangent and normals at the extremities of latus rectum of this parabola forms a quadrilateral ABCD.

- 1.** The equation of the circle 'C' is :
 (a) $x^2 + y^2 - 2x - 31 = 0$
 (b) $x^2 + y^2 - 2y - 31 = 0$
 (c) $x^2 + y^2 - 2x - 2y - 31 = 0$
 (d) $x^2 + y^2 + 2x + 2y = 31$
- 2.** The given parabola is equal to which of the following parabola ?
 (a) $y^2 = 16\sqrt{2}x$
 (b) $x^2 = -4\sqrt{2}y$
 (c) $y^2 = -\sqrt{2}x$
 (d) $y^2 = 8\sqrt{2}x$
- 3.** The area of the quadrilateral ABCD is :
 (a) 16
 (b) 8
 (c) 64
 (d) 32

Answers

1.	(a)	2.	(d)	3.	(c)							
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Exercise-4 : Matching Type Problems

1.

Column-I		Column-II	
(A)	The equation of tangent to the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ which cuts off equal intercepts on axes is $x - y = a$ where $ a $ equal to	(P)	$\sqrt{2}$
(B)	The normal $y = mx - 2am - am^2$ to the parabola $y^2 = 4ax$ subtends a right angle at the vertex if $ m $ equal to	(Q)	$\sqrt{3}$
(C)	The equation of the common tangent to parabola $y^2 = 4x$ and $x^2 = 4y$ is $x + y + \frac{k}{\sqrt{3}} = 0$, then k is equal to	(R)	$\sqrt{8}$
(D)	An equation of common tangent to parabola $y^2 = 8x$ and the hyperbola $3x^2 - y^2 = 3$ is $4x - 2y + \frac{k}{\sqrt{2}} = 0$, then k is equal to	(S)	$\sqrt{41}$
		(T)	2

2.

Column-I		Column-II	
(A)	Area of ΔPQR is equal to	(P)	2
(B)	Radius of circumcircle of ΔPQR is equal to	(Q)	$\frac{5}{2}$
(C)	Distance of the vertex from the centroid of ΔPQR is equal to	(R)	$\frac{3}{2}$
(D)	Distance of the centroid from the circumcentre of ΔPQR is equal to	(S)	$\frac{2}{3}$
		(T)	$\frac{11}{6}$

Answers

1. A $\rightarrow$ S; B $\rightarrow$ P; C $\rightarrow$ Q; D $\rightarrow$ R2. A $\rightarrow$ P; B $\rightarrow$ Q; C $\rightarrow$ S; D $\rightarrow$ T

Exercise-1 : Single Choice Problems

1. If CF be the perpendicular from the centre C of the ellipse $\frac{x^2}{12} + \frac{y^2}{8} = 1$, on the tangent at any point P and G is the point where the normal at P meets the major axis, then the value of $(CF \cdot PG)$ equals to :
 (a) 5 (b) 6 (c) 8 (d) None of these
2. The minimum length of intercept on any tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ cut by the circle $x^2 + y^2 = 25$ is :
 (a) 8 (b) 9 (c) 2 (d) 11
3. The point on the ellipse $x^2 + 2y^2 = 6$, whose distance from the line $x + y = 7$ is minimum is :
 (a) (2, 3) (b) (2, 1) (c) (1, 0) (d) None of these
4. If lines $2x + 3y = 10$ and $2x - 3y = 10$ are tangents at the extremities of a latus rectum of an ellipse; whose centre is origin, then the length of the latus rectum is :
 (a) $\frac{110}{27}$ (b) $\frac{98}{27}$ (c) $\frac{100}{27}$ (d) $\frac{120}{27}$
5. The area bounded by the circle $x^2 + y^2 = a^2$ and the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is equal to the area of another ellipse having semi-axes :
 (a) $a + b$ and b (b) $a - b$ and a (c) a and b (d) None of these
6. If F_1 and F_2 are the feet of the perpendiculars from foci S_1 and S_2 of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ on the tangent at any point P of the ellipse, then :
 (a) $S_1F_1 + S_2F_2 \geq 2$ (b) $S_1F_1 + S_2F_2 \geq 3$ (c) $S_1F_1 + S_2F_2 \geq 6$ (d) $S_1F_1 + S_2F_2 \geq 8$

7. Consider the ellipse $\frac{x^2}{f(k^2 + 2k + 5)} + \frac{y^2}{f(k + 11)} = 1$, where $f(x)$ is a positive decreasing function, then the value of k for which major axis coincides with x -axis is :
 (a) $k \in (-7, -5)$ (b) $k \in (-5, -3)$ (c) $k \in (-3, 2)$ (d) None of these
8. If area of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ inscribed in a square of side length $5\sqrt{2}$ is A , then $\frac{A}{\pi}$ equals to :
 (a) 12 (b) 10 (c) 8 (d) 11
9. Any chord of the conic $x^2 + y^2 + xy = 1$ passing through origin is bisected at a point (p, q) , then $(p + q + 12)$ equals to :
 (a) 13 (b) 14 (c) 11 (d) 12
10. Tangents are drawn from the point $(4, 2)$ to the curve $x^2 + 9y^2 = 9$, the tangent of angle between the tangents :
 (a) $\frac{3\sqrt{3}}{5\sqrt{17}}$ (b) $\frac{\sqrt{43}}{10}$ (c) $\frac{\sqrt{43}}{5}$ (d) $\sqrt{\frac{3}{17}}$

Answers

1.	(c)	2.	(a)	3.	(b)	4.	(c)	5.	(b)	6.	(d)	7.	(c)	8.	(a)	9.	(d)	10.	(c)
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Paragraph for Question Nos. 1 to 2

An ellipse has semi-major axis of length 2 and semi-minor axis of length 1. It slides between the co-ordinate axes in the first quadrant, while maintaining contact with both x-axis and y-axis.

An ellipse has semi-major axis of length 2 and semi-minor axis of length 1. It slides between the co-ordinate axes in the first quadrant, while maintaining contact with both x-axis and y-axis.

- (d) $(x-2)^2 + (y-1)^2 = 3$

- (d) $x^2 - y^2 + \frac{1}{x^2} - \frac{1}{y^2} = 2\sqrt{3} + 4$

Paragraph for Question Nos. 3 to 5

A coplanar beam of light emerging from a point source have the equation $\lambda x - y + 2(1 + \lambda) = 0$, $\forall \lambda \in \mathbb{R}$; the rays of the beam strike an elliptical surface and get reflected inside the ellipse. The reflected rays form another convergent beam having the equation $\mu x - y + 2(1 - \mu) = 0$, $\forall \mu \in \mathbb{R}$. Further it is found that the foot of the perpendicular from the point $(2, 2)$ upon any tangent to the ellipse lies on the circle $x^2 + y^2 - 4y - 5 = 0$

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- (d) $\frac{1}{2}$

- (d) $2\sqrt{5}$

- (d) $2\sqrt{5}$

| **Answers** |

1.	(b)	2.	(a)	3.	(c)	4.	(d)	5.	(a)						
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Exercise-3 : Matching Type Problems

1.

Column-I		Column-II	
(A)	If the tangent to the ellipse $x^2 + 4y^2 = 16$ at the point $P(4 \cos \phi, 2 \sin \phi)$ is a normal to the circle $x^2 + y^2 - 8x - 4y = 0$ then $\frac{\phi}{2}$ may be	(P)	0
(B)	The eccentric angle(s) of a point on the ellipse $x^2 + 3y^2 = 6$ at a distance 2 units from the centre of the ellipse is/are	(Q)	$\cos^{-1}\left(-\frac{2}{3}\right)$
(C)	The eccentric angle of point of intersection of the ellipse $x^2 + 4y^2 = 4$ and the parabola $x^2 + 1 = y$ is	(R)	$\frac{\pi}{4}$
(D)	If the normal at the point $P(\sqrt{14} \cos \theta, \sqrt{5} \sin \theta)$ to the ellipse $\frac{x^2}{14} + \frac{y^2}{5} = 1$ intersect it again at the point $Q(\sqrt{14} \cos 2\theta, \sqrt{5} \sin 2\theta)$, then θ is	(S)	$\frac{5\pi}{4}$
		(T)	$\frac{\pi}{2}$

Answers

1. A $\rightarrow$ P, R; B $\rightarrow$ R, S; C $\rightarrow$ P; D $\rightarrow$ Q

Exercise-4 : Subjective Type Problems

1. For the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. Let O be the centre and S and S' be the foci. For any point P on the ellipse the value of $PS \cdot PS'$ (where d is the distance of O from the tangent at P) is equal to
2. Number of perpendicular tangents that can be drawn on the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ from point $(6, 7)$ is

Answers

1.	4	2.	0										
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Exercise-1 : Single Choice Problems

- The normal to curve $xy = 4$ at the point $(1, 4)$ meets the curve again at :
 (a) $(-4, -1)$ (b) $\left(-8, -\frac{1}{2}\right)$ (c) $\left(-16, -\frac{1}{4}\right)$ (d) $(-1, -4)$
- Let $PQ : 2x + y + 6 = 0$ is a chord of the curve $x^2 - 4y^2 = 4$. Coordinates of the point $R(\alpha, \beta)$ that satisfy $\alpha^2 + \beta^2 - 1 \leq 0$; such that area of triangle PQR is minimum; are given by :
 (a) $\left(\frac{-2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$ (b) $\left(\frac{-2}{\sqrt{5}}, \frac{-1}{\sqrt{5}}\right)$
 (c) $\left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$ (d) $\left(\frac{2}{\sqrt{5}}, \frac{-1}{\sqrt{5}}\right)$
- If $y = mx + c$ be a tangent to hyperbola $\frac{x^2}{\lambda^2} - \frac{y^2}{(\lambda^3 + \lambda^2 + \lambda)^2} = 1$, then least value of $16m^2$ equals to :
 (a) 0 (b) 1 (c) 4 (d) 9
- Let the double ordinate PP' of the hyperbola $\frac{x^2}{4} - \frac{y^2}{3} = 1$ is produced both sides to meet asymptotes of hyperbola in Q and Q' . The product $(PQ)(PQ')$ is equal to :
 (a) 3 (b) 4 (c) 1 (d) 5
- If eccentricity of conjugate hyperbola of the given hyperbola :

$$|\sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-5)^2}| = 3$$
 is e' , then value of $8e'$ is :
 (a) 12 (b) 14 (c) 17 (d) 10

6. A normal to the hyperbola $\frac{x^2}{4} - \frac{y^2}{1} = 1$ has equal intercepts on positive x and positive y -axes. If this normal touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then $3(a^2 + b^2)$ is equal to :
- (a) 5 (b) 25 (c) 16 (d) None of these
7. Locus of a point, whose chord of contact with respect to the circle $x^2 + y^2 = 4$ is a tangent to the hyperbola $xy = 1$ is a/an :
- (a) ellipse (b) circle
(c) hyperbola (d) parabola
8. Let the chord $x \cos \alpha + y \sin \alpha = p$ of the hyperbola $\frac{x^2}{16} - \frac{y^2}{18} = 1$ subtends a right angle at the centre. Let diameter of the circle, concentric with the hyperbola, to which the given chord is a tangent is d , then $\frac{d}{4}$ is equal to :
- (a) 4 (b) 5 (c) 6 (d) 7
9. If the tangent and normal at a point on rectangular hyperbola cut-off intercept a_1, a_2 on x -axis and b_1, b_2 on the y -axis, then $a_1 a_2 + b_1 b_2$ is equal to :
- (a) 2 (b) $\frac{1}{2}$ (c) 0 (d) -1

Answers

1.	(c)	2.	(b)	3.	(d)	4.	(a)	5.	(d)	6.	(b)	7.	(c)	8.	(c)	9.	(c)		
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Exercise-2 : One or More than One Answer is/are Correct

1. A common tangent to the hyperbola $9x^2 - 16y^2 = 144$ and the circle $x^2 + y^2 = 9$ is/are :
- (a) $y = \frac{3}{\sqrt{7}}x + \frac{15}{\sqrt{7}}$ (b) $y = 3\sqrt{\frac{2}{\sqrt{7}}}x + \frac{25}{\sqrt{7}}$
 (c) $y = 2\sqrt{\frac{3}{7}}x + 15\sqrt{7}$ (d) $y = -3\sqrt{\frac{2}{\sqrt{7}}}x + \frac{25}{\sqrt{7}}$
2. Tangents are drawn to the hyperbola $x^2 - y^2 = 3$ which are parallel to the line $2x + y + 8 = 0$. Then their points of contact is/are :
- (a) (2, 1) (b) (2, -1)
 (c) (-2, -1) (d) (-2, 1)
3. If the line $ax + by + c = 0$ is normal to the curve $xy = 1$, then:
- (a) $a > 0, b > 0$ (b) $a > 0, b < 0$
 (c) $b < 0, a < 0$ (d) $a < 0, b > 0$
4. A circle cuts rectangular hyperbola $xy = 1$ in the points (x_r, y_r) , $r = 1, 2, 3, 4$ then :
- (a) $y_1y_2y_3y_4 = 1$ (b) $x_1x_2x_3x_4 = 1$
 (c) $x_1x_2x_3x_4 = y_1y_2y_3y_4 = -1$ (d) $y_1y_2y_3y_4 = 0$

Answers

1.	(b, d)	2.	(b, d)	3.	(b, d)	4.	(a, b)				
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Paragraph for Question Nos. 1 to 3

A point P moves such that sum of the slopes of the normals drawn from it to the hyperbola $xy = 16$ is equal to the sum of the ordinates of the feet of the normals. Let ' P ' lies on the curve C , then :

A point P moves such that sum of the slopes of the normals drawn from it to the hyperbola $xy = 16$ is equal to the sum of the ordinates of the feet of the normals. Let ' P ' lies on the curve C , then :

1. The equation of 'C' is :
- (a) $x^2 = 4y$ (b) $x^2 = 16y$
(c) $x^2 = 12y$ (d) $y^2 = 8x$
2. If tangents are drawn to the curve C, then the locus of the midpoint of the portion of tangent intercepted between the co-ordinate axes, is :
- (a) $x^2 = 4y$ (b) $x^2 = 2y$
(c) $x^2 + 2y = 0$ (d) $x^2 + 4y = 0$
3. Area of the equilateral triangle, inscribed in the curve C, and having one vertex same as the vertex of C is :
- (a) $768\sqrt{3}$ (b) $776\sqrt{3}$
(c) $760\sqrt{3}$ (d) None of these

Answers

1.	(b)	2.	(c)	3.	(a)
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Exercise-4 : Subjective Type Problems

1. Let $y = mx + c$ be a common tangent to $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and $\frac{x^2}{4} + \frac{y^2}{3} = 1$, then find the value of $m^2 + c^2$.
2. The maximum number of normals that can be drawn to an ellipse/hyperbola passing through a given point is :
3. Tangent at P to rectangular hyperbola $xy = 2$ meets coordinate axes at A and B , then area of triangle OAB (where O is origin) is :

Answers

1.	8	2.	4	3.	4														
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Trigonometry

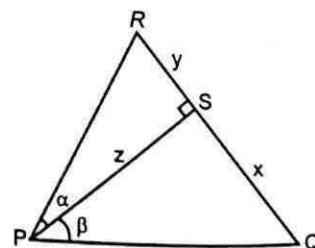
- 22.** Compound Angles
- 23.** Trigonometric Equations
- 24.** Solution of Triangles
- 25.** Inverse Trigonometric Functions

COMPOUND ANGLES

Exercise-1 : Single Choice Problems

- $\left(\cos^4 \frac{\pi}{24} - \sin^4 \frac{\pi}{24} \right)$ equals :
 - $\frac{1}{\sqrt{2}}$
 - $\frac{\sqrt{6}-\sqrt{2}}{4}$
 - $\frac{\sqrt{6}+\sqrt{2}}{4}$
 - $\frac{\sqrt{3}+1}{2}$
- If $a \sin x + b \cos(c+x) + b \cos(c-x) = \alpha$, $\alpha > a$, then the minimum value of $|\cos c|$ is :
 - $\sqrt{\frac{\alpha^2 - a^2}{b^2}}$
 - $\sqrt{\frac{\alpha^2 - a^2}{2b^2}}$
 - $\sqrt{\frac{\alpha^2 - a^2}{3b^2}}$
 - $\sqrt{\frac{\alpha^2 - a^2}{4b^2}}$
- If all values of $x \in (a, b)$ satisfy the inequality $\tan x \tan 3x < -1$, $x \in \left(0, \frac{\pi}{2}\right)$, then the maximum value $(b-a)$ is :
 - $\frac{\pi}{12}$
 - $\frac{\pi}{3}$
 - $\frac{\pi}{6}$
 - $\frac{\pi}{4}$
- $\sum_{r=1}^8 \tan(rA) \tan((r+1)A)$ where $A = 36^\circ$ is :
 - $-10 - \tan A$
 - $-10 + \tan A$
 - -10
 - -9
- Let $f(x) = 2 \operatorname{cosec} 2x + \sec x + \operatorname{cosec} x$, then minimum value of $f(x)$ for $x \in \left(0, \frac{\pi}{2}\right)$ is :
 - $\frac{1}{\sqrt{2}-1}$
 - $\frac{2}{\sqrt{2}-1}$
 - $\frac{1}{\sqrt{2}+1}$
 - $\frac{2}{\sqrt{2}+1}$
- The exact value of $\operatorname{cosec} 10^\circ + \operatorname{cosec} 50^\circ - \operatorname{cosec} 70^\circ$ is :
 - 4
 - 5
 - 6
 - 8
- If $u = \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} + \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$, then the difference between the maximum and minimum values of u^2 is given by :
 - $2(a^2 + b^2)$
 - $2\sqrt{a^2 + b^2}$
 - $(a+b)^2$
 - $(a-b)^2$

8. If $u_n = \sin(n\theta) \sec^n \theta$, $v_n = \cos(n\theta) \sec^n \theta$, $n \in N$, $n \neq 1$, then $\frac{v_n - v_{n-1}}{u_{n-1}} + \frac{1}{n} \frac{u_n}{v_n} =$
- (a) $-\cot \theta + \frac{1}{n} \tan(n\theta)$ (b) $\cot \theta + \frac{1}{n} \tan(n\theta)$
 (c) $\tan \theta + \frac{1}{n} \tan(n\theta)$ (d) $-\tan \theta + \frac{\tan(n\theta)}{n}$
9. If $a \cos^2 3\alpha + b \cos^4 \alpha = 16 \cos^6 \alpha + 9 \cos^2 \alpha$ is an identity, then
- (a) $a = 1, b = 24$ (b) $a = 3, b = 24$ (c) $a = 4, b = 2$ (d) $a = 7, b = 18$
10. Maximum value of $\cos x (\sin x + \cos x)$ is equal to :
- (a) $\sqrt{2}$ (b) 2 (c) $\frac{\sqrt{2}+1}{2}$ (d) $\sqrt{2}+1$
11. If $\frac{\sin A}{\sin B} = \frac{\sqrt{3}}{2}$ and $\frac{\cos A}{\cos B} = \frac{\sqrt{5}}{2}$, $0 < A, B < \frac{\pi}{2}$ then $\tan A + \tan B$ is equal to :
- (a) $\sqrt{\frac{3}{5}}$ (b) $\sqrt{\frac{5}{3}}$ (c) $\frac{\sqrt{3}+\sqrt{5}}{\sqrt{5}}$ (d) $\frac{\sqrt{3}+\sqrt{5}}{\sqrt{3}}$
12. Let $0 \leq \alpha, \beta, \gamma, \delta \leq \pi$ where β and γ are not complementary such that
- $$2 \cos \alpha + 6 \cos \beta + 7 \cos \gamma + 9 \cos \delta = 0$$
- and $2 \sin \alpha - 6 \sin \beta + 7 \sin \gamma - 9 \sin \delta = 0$
- If $\frac{\cos(\alpha + \delta)}{\cos(\beta + \gamma)} = \frac{m}{n}$ where m and n are relatively prime positive numbers, then the value of $(m + n)$ is equal to :
- (a) 11 (b) 10 (c) 9 (d) 7
13. If $-\pi < \theta < -\frac{\pi}{2}$, then $\left| \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} + \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}} \right|$ is equal to :
- (a) $2 \sec \theta$ (b) $-2 \sec \theta$ (c) $2 \sec \frac{\theta}{2}$ (d) $-\sec \frac{\theta}{2}$
14. If $A = \sum_{r=1}^3 \cos \frac{2r\pi}{7}$ and $B = \sum_{r=1}^3 \cos \frac{2^r \pi}{7}$, then :
- (a) $A + B = 0$ (b) $2A + B = 0$ (c) $A + 2B = 0$ (d) $A = B$
15. In a ΔPQR (as shown in figure) if $x:y:z = 2:3:6$, then the value of $\angle QPR$ is :
- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$



16. If $A = \sum_{r=1}^3 \cos \frac{2r\pi}{7}$ and $B = \sum_{r=1}^3 \cos \frac{2^r\pi}{7}$, then :
- (a) $A + B = 0$ (b) $2A + B = 0$ (c) $A + 2B = 0$ (d) $A - B = 0$
17. Let $f(x) = \sin x + 2\cos^2 x$; $\frac{\pi}{6} \leq x \leq \frac{2\pi}{3}$, then maximum value of $f(x)$ is :
- (a) 1 (b) $\frac{3}{2}$ (c) 2 (d) $\frac{5}{2}$
18. In $\triangle ABC$, $\angle C = \frac{2\pi}{3}$ then the value of $\cos^2 A + \cos^2 B - \cos A \cdot \cos B$ is equal to :
- (a) $\frac{3}{4}$ (b) $\frac{3}{2}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$
19. The number of solutions of the equation $4\sin^2 x + \tan^2 x + \cot^2 x + \operatorname{cosec}^2 x = 6$ in $[0, 2\pi]$:
- (a) 1 (b) 2 (c) 3 (d) 4
20. If $\sin A$, $\cos A$ and $\tan A$ are in G.P., then $\cos^3 A + \cos^2 A$ is equal to :
- (a) 1 (b) 2 (c) 4 (d) none
21. Range of function $f(x) = \sin\left(x + \frac{\pi}{6}\right) + \cos\left(x - \frac{\pi}{6}\right)$ is :
- (a) $[-\sqrt{2}, \sqrt{2}]$ (b) $[-\sqrt{2}(\sqrt{3}+1), \sqrt{2}(\sqrt{3}+1)]$
 (c) $\left[-\frac{\sqrt{3}+1}{\sqrt{2}}, \frac{\sqrt{3}+1}{\sqrt{2}}\right]$ (d) $\left[-\frac{\sqrt{3}-1}{\sqrt{2}}, \frac{\sqrt{3}-1}{\sqrt{2}}\right]$
22. The value of $\tan(\log_2 6) \cdot \tan(\log_2 3) \cdot \tan 1$ is always equal to :
- (a) $\tan(\log_2 6) + \tan(\log_2 3) + \tan 1$ (b) $\tan(\log_2 6) - \tan(\log_2 3) - \tan 1$
 (c) $\tan(\log_2 6) - \tan(\log_2 3) + \tan 1$ (d) $\tan(\log_2 6) + \tan(\log_2 3) - \tan 1$
23. In a triangle ABC , side $BC = 3$, $AC = 4$ and $AB = 5$. The value of $\sin A + \sin 2B + \sin 3C$ is equal to:
- (a) $\frac{24}{25}$ (b) $\frac{14}{25}$ (c) $\frac{64}{25}$ (d) none
24. If $A + B + C = 180^\circ$, then $\frac{\cos A \cos C + \cos(A+B) \cos(B+C)}{\cos A \sin C - \sin(A+B) \cos(B+C)}$ simplifies to :
- (a) $-\cot C$ (b) 0 (c) $\tan C$ (d) $\cot C$
25. If $\alpha + \gamma = 2\beta$ then the expression $\frac{\sin \alpha - \sin \gamma}{\cos \gamma - \cos \alpha}$ simplifies to :
- (a) $\tan \beta$ (b) $-\tan \beta$ (c) $\cot \beta$ (d) $-\cot \beta$

26. The product $\left(\cos \frac{x}{2}\right) \cdot \left(\cos \frac{x}{4}\right) \cdot \left(\cos \frac{x}{8}\right) \cdots \left(\cos \frac{x}{256}\right)$ is equal to :
- (a) $\frac{\sin x}{128 \sin \frac{x}{256}}$ (b) $\frac{\sin x}{256 \sin \frac{x}{256}}$ (c) $\frac{\sin x}{128 \sin \frac{x}{128}}$ (d) $\frac{\sin x}{512 \sin \frac{x}{512}}$
27. The value of the expression $\frac{\sin 7\alpha + 6 \sin 5\alpha + 17 \sin 3\alpha + 12 \sin \alpha}{\sin 6\alpha + 5 \sin 4\alpha + 12 \sin 2\alpha}$, where $\alpha = \frac{\pi}{5}$ is equal to :
- (a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{\sqrt{5}+1}{4}$ (c) $\frac{\sqrt{5}+1}{2}$ (d) $\frac{\sqrt{5}-1}{2}$
28. In a triangle ABC if $\sum \tan^2 A = \sum \tan A \tan B$, then largest angle of the triangle in radian will be :
- (a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{3\pi}{4}$
29. Which one of the following values is not the solution of the equation $\log_{|\sin x|}(|\cos x|) + \log_{|\cos x|}(|\sin x|) = 2$
- (a) $\frac{7\pi}{4}$ (b) $\frac{11\pi}{4}$ (c) $\frac{3\pi}{4}$ (d) $\frac{3\pi}{8}$
30. Range of $f(x) = \sin^6 x + \cos^6 x$ is :
- (a) $\left[\frac{1}{4}, 1\right]$ (b) $\left[\frac{1}{4}, \frac{3}{4}\right]$ (c) $\left[\frac{3}{4}, 1\right]$ (d) $[1, 2]$
31. If $y = \frac{2 \sin \alpha}{1 + \cos \alpha + \sin \alpha}$, then $\frac{1 - \cos \alpha + \sin \alpha}{1 + \sin \alpha}$ is equal to :
- (a) $\frac{1}{y}$ (b) y (c) $1 - y$ (d) $1 + y$
32. If $\frac{\tan^3 A}{1 + \tan^2 A} + \frac{\cot^3 A}{1 + \cot^2 A} = p \sec A \operatorname{cosec} A + q \sin A \cos A$, then :
- (a) $p = 2, q = 1$ (b) $p = 1, q = 2$ (c) $p = 1, q = -2$ (d) $p = 2, q = -1$
33. If θ lies in the second quadrant. Then the value of $\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} + \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}}$ is equal to :
- (a) $2 \sec \theta$ (b) $-2 \sec \theta$ (c) $2 \operatorname{cosec} \theta$ (d) 2
34. If $y = (\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2$, then minimum value of y is :
- (a) 7 (b) 8 (c) 9 (d) none of these
35. If $\log_3 \sin x - \log_3 \cos x - \log_3 (1 - \tan x) - \log_3 (1 + \tan x) = -1$, then $\tan 2x$ is equal to (wherever defined)
- (a) -2 (b) $\frac{3}{2}$ (c) $\frac{2}{3}$ (d) 6

36. If $\sin \theta + \operatorname{cosec} \theta = 2$, then the value of $\sin^8 \theta + \operatorname{cosec}^8 \theta$ is equal to :
 (a) 2 (b) 2^4 (c) 2^8 (d) more than 2^8
37. If $\tan^3 \theta + \cot^3 \theta = 52$, then the value of $\tan^2 \theta + \cot^2 \theta$ is equal to :
 (a) 14 (b) 15
 (c) 16 (d) 17
38. The maximum value of $\log_{20}(3 \sin x - 4 \cos x + 15)$ is equal to :
 (a) 1 (b) 2 (c) 3 (d) 4
39. If $x^2 + y^2 = 9$ and $4a^2 + 9b^2 = 16$, then maximum value of $4a^2x^2 + 9b^2y^2 - 12abxy$ is :
 (a) 81 (b) 100 (c) 121 (d) 144
40. If $A = \sqrt{\sin 2 - \sin \sqrt{3}}$, $B = \sqrt{\cos 2 - \cos \sqrt{3}}$, then which of the following statement is true ?
 (a) A and B both are real numbers and $A > B$
 (b) A and B both are real numbers and $A < B$
 (c) Exactly one of A and B is not real number
 (d) Both A and B are not real numbers
41. The number of real values of x such that
 $(2^x + 2^{-x} - 2 \cos x)(3^{x+\pi} + 3^{-x-\pi} + 2 \cos x)(5^{\pi-x} + 5^{x-\pi} - 2 \cos x) = 0$ is :
 (a) 1 (b) 2 (c) 3 (d) infinite
42. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has :
 (a) infinite number of real roots (b) no real roots
 (c) exactly one real root (d) exactly four real roots
43. If $\pi < \alpha < \frac{3\pi}{2}$, then the expression $\sqrt{4 \sin^4 \alpha + \sin^2 2\alpha} + 4 \cos^2 \left(\frac{\pi}{4} - \frac{\alpha}{2} \right)$ is equal to :
 (a) $2 + 4 \sin \alpha$ (b) $2 - 4 \cos \alpha$ (c) 2 (d) $2 - 4 \sin \alpha$
44. $\left(\cos \frac{\pi}{12} - \sin \frac{\pi}{12} \right) \left(\tan \frac{\pi}{12} + \cot \frac{\pi}{12} \right) =$
 (a) $\frac{1}{\sqrt{2}}$ (b) $4\sqrt{2}$ (c) $\sqrt{2}$ (d) $2\sqrt{2}$
45. $\tan(100^\circ) + \tan(125^\circ) + \tan(100^\circ) \tan(125^\circ) =$
 (a) 0 (b) $\frac{1}{2}$ (c) -1 (d) 1
46. If $\sin x + \sin^2 x = 1$, then $\cos^8 x + 2 \cos^6 x + \cos^4 x =$
 (a) 2 (b) 1 (c) 3 (d) $\frac{1}{2}$
47. The maximum value of $\log_5(3x + 4y)$, if $x^2 + y^2 = 25$ is :
 (a) 1 (b) 2 (c) 3 (d) 4

48. The number of values of θ between $-\pi$ and $\frac{3\pi}{2}$ that satisfies the equation $5 \cos 2\theta + 2 \cos^2 \frac{\theta}{2} + 1 = 0$ is :
 (a) 3 (b) 4 (c) 5 (d) 6
49. Given that $\sin \beta = \frac{4}{5}$, $0 < \beta < \pi$ and $\tan \beta > 0$, then $((3 \sin(\alpha + \beta) - 4 \cos(\alpha + \beta)) \operatorname{cosec} \alpha$ is equal to:
 (a) 2 (b) 3 (c) 4 (d) 5
50. The maximum value of $\sin\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$ for $x \in \left[0, \frac{\pi}{2}\right]$ is attained at $x =$
 (a) $\frac{\pi}{12}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$
51. The values of 'a' for which the equation $\sin x (\sin x + \cos x) = a$ has a real solution are
 (a) $1 - \sqrt{2} \leq a \leq 1 + \sqrt{2}$ (b) $2 - \sqrt{3} \leq a \leq 2 + \sqrt{3}$
 (c) $0 \leq a \leq 2 + \sqrt{3}$ (d) $\frac{1 - \sqrt{2}}{2} \leq a \leq \frac{1 + \sqrt{2}}{2}$
52. The value of $\cos 12^\circ \cos 24^\circ \cos 36^\circ \cos 48^\circ \cos 60^\circ \cos 72^\circ \cos 84^\circ$ is :
 (a) $\frac{1}{64}$ (b) $\frac{1}{128}$ (c) $\frac{1}{256}$ (d) $\frac{1}{512}$
53. The ratio of the maximum value to minimum value of $2 \cos^2 \theta + \cos \theta + 1$ is :
 (a) 32 : 7 (b) 32 : 9 (c) 4 : 1 (d) 2 : 1
54. If all values of $x \in (a, b)$ satisfy the inequality $\tan x \tan 3x < -1$, $x \in \left(0, \frac{\pi}{2}\right)$, then the maximum value $(b - a)$ is :
 (a) $\frac{\pi}{12}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{4}$
55. If a regular polygon of 'n' sides has circum radius = R and inradius = r; then each side of polygon is :
 (a) $(R + r) \tan\left(\frac{\pi}{2n}\right)$ (b) $2(R + r) \tan\left(\frac{\pi}{2n}\right)$
 (c) $(R + r) \sin\left(\frac{\pi}{2n}\right)$ (d) $2(R + r) \cot\left(\frac{\pi}{2n}\right)$
56. The value of $\cos 12^\circ + \cos 84^\circ + \cos 156^\circ + \cos 132^\circ$ is:
 (a) $\frac{1}{8}$ (b) $-\frac{1}{2}$ (c) 1 (d) $\frac{1}{2}$
57. $\frac{\sin \theta}{\cos(3\theta)} + \frac{\sin(3\theta)}{\cos(9\theta)} + \frac{\sin(9\theta)}{\cos(27\theta)} + \frac{\sin(27\theta)}{\cos(81\theta)} =$

- (a) $\frac{\sin(81\theta)}{2\cos(80\theta)\cos\theta}$ (b) $\frac{\sin(80\theta)}{2\cos(81\theta)\cos\theta}$
 (c) $\frac{\sin(81\theta)}{\cos(80\theta)\cos\theta}$ (d) $\frac{\sin(80\theta)}{\cos(81\theta)\cos\theta}$
58. The value of $\left(\sin\frac{\pi}{9}\right)\left(4 + \sec\frac{\pi}{9}\right)$ is :
 (a) $\frac{1}{2}$ (b) $\sqrt{2}$ (c) 1 (d) $\sqrt{3}$
59. If $\frac{dy}{dx} = \sin\left(\frac{x\pi}{2}\right)\cos(x\pi)$, then y is strictly increasing in :
 (a) (3, 4) (b) $\left(\frac{5}{2}, \frac{7}{2}\right)$ (c) (2, 3) (d) $\left(\frac{1}{2}, \frac{3}{2}\right)$
60. Smallest positive value of θ satisfying the equation $8\sin\theta\cos2\theta\sin3\theta\cos4\theta = \cos6\theta$; is :
 (a) $\frac{\pi}{18}$ (b) $\frac{\pi}{22}$ (c) $\frac{\pi}{24}$ (d) None of these
61. If an angle A of a triangle ABC is given by $3\tan A + 1 = 0$, then $\sin A$ and $\cos A$ are the roots of the equation
 (a) $10x^2 - 2\sqrt{10}x + 3 = 0$ (b) $10x^2 - 2\sqrt{10}x - 3 = 0$
 (c) $10x^2 + 2\sqrt{10}x + 3 = 0$ (d) $10x^2 + 2\sqrt{10}x - 3 = 0$
62. If θ is an acute angle and $\tan\theta = \frac{1}{\sqrt{7}}$, then the value of $\frac{\operatorname{cosec}^2\theta - \sec^2\theta}{\operatorname{cosec}^2\theta + \sec^2\theta}$ is :
 (a) $3/4$ (b) $1/2$ (c) 2 (d) $5/4$
63. If $2\cos\theta + \sin\theta = 1$, then $7\cos\theta + 6\sin\theta$ equals
 (a) 1 or 2 (b) 2 or 3 (c) 2 or 4 (d) 2 or 6
64. If $\sin\theta + \operatorname{cosec}\theta = 2$, then the value of $\sin^8\theta + \operatorname{cosec}^8\theta$ is equal to :
 (a) 2 (b) 2^4 (c) 2^8 (d) more than 2^8
65. If $\tan^3\theta + \cot^3\theta = 52$, then the value of $\tan^2\theta + \cot^2\theta$ is equal to :
 (a) 14 (b) 15 (c) 16 (d) 17
66. If $ABCD$ is a cyclic quadrilateral such that $12\tan A - 5 = 0$ and $5\cos B + 3 = 0$ then $\tan C + \tan D$ is equal to :
 (a) $\frac{21}{12}$ (b) $\frac{11}{12}$ (c) $-\frac{11}{12}$ (d) $-\frac{21}{12}$
67. If $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$ then $\sqrt{\tan^2\theta - \sin^2\theta}$ is equal to :
 (a) $\tan\theta\sin\theta$ (b) $-\tan\theta\sin\theta$ (c) $\tan\theta - \sin\theta$ (d) $\sin\theta - \tan\theta$